



Sample Application for Research Training and Career Development Funding

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Please note that these applications do not reflect the [new Fellowship review criteria](#), which are effective January 25, 2025, as well as other updates to application instructions. Be sure to refer to the latest [Fellowship Instructions](#) and the current Notice of Funding Opportunity when preparing an F99/K00 application.

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<https://www.nia.nih.gov/research/training/sample-applications>

PI: Yu, Kexin	Title: Examining the Longitudinal Influence of the Physical and Social Environments on Social Isolation and Cognitive Health: contextualizing the role of technology	
Received: 10/22/2019	Opportunity: RFA-AG-20-009 Clinical Trial: Not Allowed	Council: 05/2020
Competition ID: FORMS-E	FOA Title: Transition to Aging Research for Predoctoral Students (F99/K00)	
1F99AG068492-01	Dual:	Accession Number: 4366769
IPF: 7636101	Organization: UNIVERSITY OF SOUTHERN CALIFORNIA	
Former Number:	Department: Social Work	
IRG/SRG: ZAG1 ZIJ-1 (M2)	AIDS: N	Expedited: N
<u>Subtotal Direct Costs</u> (excludes consortium F&A) Year 1: █	Animals: N Humans: Y Clinical Trial: N Current HS Code: 30 HESC: N HFT: N	New Investigator: Early Stage Investigator:
<i>Senior/Key Personnel:</i>	<i>Organization:</i>	<i>Role Category:</i>
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Lawrence Palinkas	University of Southern California	Other (Specify)-Co-Sponsor
Iris Chi	University of Southern California	Other (Specify)-Sponsor

Reference Letters

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█	█	█

Additions for Review

Accepted Publication	Notification of accepted publications
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APPLICATION FOR FEDERAL ASSISTANCE

SF 424 (R&R)

3. DATE RECEIVED BY STATE		State Application Identifier
1. TYPE OF SUBMISSION*		4.a. Federal Identifier
☐ Pre-application ☐ Application ☒ Changed/Corrected Application		b. Agency Routing Number
2. DATE SUBMITTED 2019-10-22	Application Identifier	c. Previous Grants.gov Tracking Number
5. APPLICANT INFORMATION		Organizational DUNS*: [REDACTED]
Legal Name*: University of Southern California		
Department:		
Division:		
Street1*: Department of Contracts and Grants		
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Phone Number*: [REDACTED]		Fax Number: [REDACTED]
Email: [REDACTED]		
6. EMPLOYER IDENTIFICATION NUMBER (EIN) or (TIN)* [REDACTED]		
7. TYPE OF APPLICANT*		O: Private Institution of Higher Education
Other (Specify):		
☒ Small Business Organization Type ☐ Women Owned ☐ Socially and Economically Disadvantaged		
8. TYPE OF APPLICATION*		If Revision, mark appropriate box(es).
☒ New ☐ Resubmission		☐ A. Increase Award ☐ B. Decrease Award ☐ C. Increase Duration
☐ Renewal ☐ Continuation ☐ Revision		☐ D. Decrease Duration ☐ E. Other (specify) :
Is this application being submitted to other agencies?* ☐ Yes ☒ No What other Agencies?		
9. NAME OF FEDERAL AGENCY* National Institutes of Health		10. CATALOG OF FEDERAL DOMESTIC ASSISTANCE NUMBER TITLE:
11. DESCRIPTIVE TITLE OF APPLICANT'S PROJECT* Examining the Longitudinal Influence of the Physical and Social Environments on Social Isolation and Cognitive Health: contextualizing the role of technology		
12. PROPOSED PROJECT Start Date* Ending Date* 07/01/2020 06/30/2025		13. CONGRESSIONAL DISTRICTS OF APPLICANT CA-037

SF 424 (R&R) APPLICATION FOR FEDERAL ASSISTANCE**Page 2****14. PROJECT DIRECTOR/PRINCIPAL INVESTIGATOR CONTACT INFORMATION**

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15. ESTIMATED PROJECT FUNDING

- a. Total Federal Funds Requested* [REDACTED]
- b. Total Non-Federal Funds* [REDACTED]
- c. Total Federal & Non-Federal Funds* [REDACTED]
- d. Estimated Program Income* [REDACTED]

16. IS APPLICATION SUBJECT TO REVIEW BY STATE EXECUTIVE ORDER 12372 PROCESS?*

- a. YES ☐ THIS PREAPPLICATION/APPLICATION WAS MADE AVAILABLE TO THE STATE EXECUTIVE ORDER 12372 PROCESS FOR REVIEW ON:
- DATE:
- b. NO ☒ PROGRAM IS NOT COVERED BY E.O. 12372; OR
- ☐ PROGRAM HAS NOT BEEN SELECTED BY STATE FOR REVIEW

17. By signing this application, I certify (1) to the statements contained in the list of certifications* and (2) that the statements herein are true, complete and accurate to the best of my knowledge. I also provide the required assurances * and agree to comply with any resulting terms if I accept an award. I am aware that any false, fictitious, or fraudulent statements or claims may subject me to criminal, civil, or administrative penalties. (U.S. Code, Title 18, Section 1001)

☒ I agree*

* The list of certifications and assurances, or an Internet site where you may obtain this list, is contained in the announcement or agency specific instructions.

18. SFLL or OTHER EXPLANATORY DOCUMENTATION

File Name:

19. AUTHORIZED REPRESENTATIVE

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Phone Number*: [REDACTED] Fax Number: [REDACTED] Email*: [REDACTED]

Signature of Authorized Representative*

Steven L. Misuraca

Date Signed*

10/22/2019

20. PRE-APPLICATION File Name:**21. COVER LETTER ATTACHMENT** File Name: 02_cover_letter1016533154.pdf

424 R&R and PHS-398 Specific

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Project/Performance Site Location(s)**Project/Performance Site Primary Location**

☐ I am submitting an application as an individual, and not on behalf of a company, state, local or tribal government, academia, or other type of organization.

Organization Name: University of Southern California
Duns Number: [REDACTED]
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Street2:
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County: [REDACTED]
State*: [REDACTED]
Province:
Country*: [REDACTED]
Zip / Postal Code*: [REDACTED]
Project/Performance Site Congressional District*: CA-037

Additional Location(s)

File Name:

RESEARCH & RELATED Other Project Information

1. Are Human Subjects Involved?* ☒ Yes ☐ No 1.a. If YES to Human Subjects Is the Project Exempt from Federal regulations? ☐ Yes ☒ No If YES, check appropriate exemption number: — 1 — 2 — 3 — 4 — 5 — 6 — 7 — 8 If NO, is the IRB review Pending? ☒ Yes ☐ No IRB Approval Date: Human Subject Assurance Number 00007099	
2. Are Vertebrate Animals Used?* ☐ Yes ☒ No 2.a. If YES to Vertebrate Animals Is the IACUC review Pending? ☐ Yes ☐ No IACUC Approval Date: Animal Welfare Assurance Number	
3. Is proprietary/privileged information included in the application?* ☐ Yes ☒ No	
4.a. Does this project have an actual or potential impact - positive or negative - on the environment?* ☐ Yes ☒ No 4.b. If yes, please explain: 4.c. If this project has an actual or potential impact on the environment, has an exemption been authorized or an environmental assessment (EA) or environmental impact statement (EIS) been performed? ☐ Yes ☐ No 4.d. If yes, please explain:	
5. Is the research performance site designated, or eligible to be designated, as a historic place?* ☐ Yes ☒ No 5.a. If yes, please explain:	
6. Does this project involve activities outside the United States or partnership with international collaborators?* ☐ Yes ☒ No 6.a. If yes, identify countries: 6.b. Optional Explanation:	
7. Project Summary/Abstract*	Filename 04__Project_Summary1016533226.pdf
8. Project Narrative*	05__Project_Narrative1016533227.pdf
9. Bibliography & References Cited	06__References_Cited1016533157.pdf
10. Facilities & Other Resources	Facilities_and_Other_Resource_KY1016533284.pdf
11. Equipment	No_Major_Equipment_Doc1016533159.pdf
12. Other Attachments	Certification_Letter_Kexin_Yu_Oct_191016533214.pdf

Project Summary

Social wellbeing and cognitive health are two pillars of successful aging, and they are interrelated. Three decades of research have consistently shown that social isolation is associated with declines in cognitive functioning. Although existing literature has explored the individual-level risk and protective factors of social isolation and cognitive health, researchers' attention has been less paid to how does the physical and social characteristics of the living environment affect the experience of social isolation and cognitive health. The Ecological Theory of Aging (ETA) posits environmental factors have significant influences on health and social wellbeing. Negative physical characteristics of the environment were hypothesized to related to increased social isolation and cognitive decline longitudinally. Perceptions of social environments are often affected by the physical environment. Hence, it is hypothesized that the social environment mediated the longitudinal association between the physical environment and social isolation as well as cognitive decline.

An increasing amount of research investigated the association between ICT use and social isolation, yet whether ICT use would interact with the effect of environmental factors on the risk of social isolation has not yet been empirically tested. Besides, the influence of ICT use on cognitive health has not been well understood. Promoting social interaction using technology-based approaches have the potential to alleviate social isolation among older people and prevent cognitive decline at the same time.

The proposed dissertation work (F99) adopts a mixed-methods approach. National Health & Aging Trend Study (NHATS) waves 5 to 8 data will be used to investigate the hypothesized relationships between physical and social environment, ICT use, social isolation, and cognitive decline with latent growth curve modeling. To explain how and why social and physical aspects of the living environment affect the social wellbeing and health outcomes of its older residents, focus groups will be conducted to elicit the older adults' perception of the environmental influence on social wellbeing. At the K00 phase, the PI will use the longitudinal data from the ongoing randomized controlled trial "Internet-based Conversational Engagement Clinical Trial (I-CONNECT)" to investigate the effectiveness of an ICT-enhanced conversational intervention on social isolation and its effect on cognitive health among older adults with normal cognition or mild cognitive impairment (MCI).

The proposed research will produce knowledge on the influence of environmental factors on social isolation and cognitive health. Knowledge built at F99 and K00 stages will be synthesized to inform the development of a technology-facilitated intervention program for social isolation and cognitive decline from the perspective of ETA.

Project Narrative

Social isolation and cognitive health are interrelated critical outcomes in the aging process, yet their association with the physical and social living environment has not been extensively researched. Promoting social interaction using technology-based approaches have the potential to alleviate social isolation and prevent cognitive decline. The proposed work will produce longitudinal empirical evidence on the influence of environmental factors on social isolation and cognitive health, examining the effectiveness of an ICT-enhanced conversational intervention for social isolation reduction and delay the onset of dementia, and eventually, synthesizing the findings to inform the development of a technology-facilitated intervention program for social isolation and cognitive decline using an ecological approach.

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Facilities & Other Resources

USC Suzanne Dworak-Peck School of Social Work

The USC Suzanne Dworak-Peck School of Social Work (SSW) is strategically placed at the crux of critical societal problems in urban Los Angeles, a vastly diverse setting that reflects, in many ways, all modern complex urban environments. The School is home to the **Hamovitch Center for Science in the Human Services** (HRC), which was first endowed in 1988; the **Edward R. Roybal Institute on Aging**, which conducts translational research on aging in minority populations; the **Center for Innovation and Research on Veterans & Military Families**, dedicated to advancing evidence-based interventions and workforce development for service members, veterans, and their families; the **National Center for School Crisis and Bereavement**, whose mission is to provide research and services to children and their families experiencing extreme trauma from school-related violence; the **USC Center for Artificial Intelligence in Society** which conducts research on innovative artificial intelligence applications' impact on society; the **Children's Data Network** that maintains a network for a California-wide integrated child services big data system specifically designed for epidemiological research on neighborhood-level effects on child development and the implementation of interventions that will provide therapeutic alternatives to the present juvenile justice system; the **National Center for Excellence in Homeless Services' Research Core** that conducts research on homelessness and housing focusing on enhancing the health and well-being of persons experiencing homelessness and on solutions to homelessness through permanent housing and supportive services; the **Center for LGBT Health Equity** that leads scientific inquiry into the physical, emotional and social health of LGBT youth, adults and family, and the **California Social Welfare Archives**, which creates and houses the most complete documentation of social welfare history in California.

The School provides more than 20,000 square feet of dedicated research space in two locations (University Park Center and the City Center) for the investigators, research project staff, and administration; secure data storage facilities; symposia spaces that include state-of-the-art video conferencing equipment; and an information technology (IT) network that contains computer stations and digital information and training laboratories offering academic and technological resources and services for students and the faculty. A dedicated IT department maintains computers and local network services. The IT department provides technical assistance to investigators and is responsible for access and security on computers, laptops, and mobile devices. Maintained by the IT department, the offices are equipped with Dell Optiplex 990 desktop computers, featuring Microsoft Office Suite, SAS 9.2 for advanced statistical analysis, NVivo for qualitative data analysis, Bluejeans teleconferencing software, and a camera and microphone. The IT department provides daily backups of each computer. These backups and the data stored on these computers are secured by the university's large-data cyber security system with appropriate firewalls.

The School maintains a strong infrastructure of staff experts in research administration, data analysis and database management. Under the direction of the Research Council, an elected faculty governance body, the Associate Dean of Research and the Director of Research Administration, research administration staff, featuring two nationally Certified Research Administrators (CRAs), provides investigators assistance with proposal development and submission, pre-award responses and budget development, post-award project and budget management, and research compliance. With the assistance of the CRA-led staff, PIs have access to Cayuse and TARA, USC's electronic research administration and information portals. These portals assist researchers and the research administration staff with various aspects of managing sponsored research projects, including pre-award and post-award activities. In addition, investigators have access to iStar, the USC online institutional review board (IRB) application system used to submit, track, and review all research studies involving human subjects. The IRB staff or IRB designee conducts an initial review of all research applications involving human subjects. The IRB staff conducts a thorough pre-review of the application to verify the correct level of review and evaluate the protocol and supporting documents (e.g., consent form, recruitment materials). Any significant changes to the approved study must be submitted and reviewed by the IRB prior to implementation of changes. For studies designated as expedited or full board, IRB review is required from a designated reviewer or the full board. Access to other USC systems relevant to the research mission include disClose for conflict of interest disclosure and management and Sophia for managing inventions, intellectual property, and patents.

A team of biostatistician, bioengineering statistician and a database manager assists investigators and doctoral students in creating databases and datasets, developing design and analytic approaches, estimating sampling size requirements for specific research questions, planning and performing interim and final analyses, and writing reports to summarize results for manuscript preparation. Additionally, the School has access to senior

biostatisticians and methodologists with expertise in specialized areas through its Southern California Statistical Support Network. PIs with big datasets and complex computational models also have access to the USC High Performance Computing Cluster's (HPCC) extensive computer facilities. HPCC features a diverse mix of computing and data resources. Two Linux clusters constitute the principal computing resource. In addition, HPCC has a central facility that provides more than 1.2 petabytes of combined disk storage and potential access to nearly a petabyte of tape storage. The HPCC computing resource consists of two shared-head nodes and various compute nodes. All PIs can obtain free accounts to supplement the HRC database management and data analysis resources to enable successful performance of statistical analyses of longitudinal "big data."

USC Institute for Creative Technologies

The USC Institute for Creative Technologies is an interdisciplinary laboratory devoted to the advancement of virtual reality training applications, and provides a unique and supportive environment for the advancement of virtual human technology. With two hundred faculty, researchers, graduate students and support staff, no other laboratory in the world has such ready access to excellent talent across the range of disciplines that impact virtual human design: computer graphics, computer animation, computer games, audio rendering, artificial intelligence, natural language processing, and entertainment industry professional content producers. Institute for Creative Technologies has long enjoyed a close relationship with the entertainment industry through its School of Cinema and Television, as well as its Entertainment Technology Center. In 1999, the University expanded its relationship with the entertainment industry when it entered into an agreement with the US Army Research Office to create a University Affiliated Research. Institute for Creative Technologies now represents a unique collaboration between the Hollywood entertainment community, the US Army, and academia, with its mission to create a world-class research facility to develop training and simulation applications, and a successful track record of transitioning technology.

Institute for Creative Technologies has three stories of dedicated space for administration, investigators, and research project staff members and students; secure data storage facilities; two symposia spaces with projector equipment as well as various smaller conference rooms. Institute for Creative Technologies has a full time IT staff that maintains computers and local network services, and a central facility that provides encrypted data storage. The IT department provides technical assistance to investigators and is responsible for access and security on computers, laptops, and mobile devices. All Institute for Creative Technologies investigators on this grant have access to these support services, as well as both desktop and laptop computers with all required software (e.g., Microsoft Office Suite, analytic software such as SPSS or SAS to programing software such as Python, software for physiological data collection such as AcqKnowledge, etc.) as well as sufficient hardware (e.g., high-end graphics cards).

Institute for Creative Technologies also has substantial facilities for conducting research with human subjects, including a 20-participant computer lab and multiple rooms for running participants in individual settings. In addition, Institute for Creative Technologies has access to on-campus facilities for conducting research with human subjects at the University of Southern California. These laboratories include single-, double- and group-occupancy configurations, and allow for easier recruitment of specialized subject populations (e.g., subjects with clinical training).

Scientific Environment

The environment provided by SSW and the different centers contributes to the probability of success of the proposed application through excellent staffing and a climate of encouragement for scientific research. Evidence-based interventions and translational science are core components of the scientific agenda and intellectual foundation of USC as a university and School. SSW has pioneered the science of social work based on a foundation of an interdisciplinary faculty that along with PhD-level social workers also features sociologists, psychologists, anthropologists, nurses, engineers, and a developmental pediatrician. The School continues to have a solid record of success in receiving competitive federal grants by virtue of its strong research environment. These interdisciplinary faculty members work with other schools at USC to produce research that has an international impact and tackles the most challenging problems of society. Through these collaborative institutional networks, the School is well situated to share the benefits of the university's impressive scientific environment.

USC is qualified as one of a few premier research institutions on which our country depends for a steady stream of new knowledge, art, and technology. USC has more than \$700 million in annual research expenditures and has ranked among the top 10 private universities in federally supported research activities. This federal

support has created a rich array of core facilities and other resources that are directed by the university's internal funding programs to be directed toward establishing a culture and infrastructure for building NIH and NSF research centers. USC is the oldest private research university in the West and one of the nation's most highly selective institutions in terms of the student body it attracts. With 17 professional schools in addition to the Dornsife College of Letters, Arts and Sciences, USC is ranked 11th among U.S. private universities in dollar volume of federal research support. The accredited IRB, libraries, and other resources of the university provide exceptional support of research and scholarship. Federal research funding has doubled in the past 10 years. USC graduates have an invaluable worldwide support network of more than 180,000 living alumni in the Trojan Family. Alumni maintain a close association with the university, mentoring students and opening career doors for graduates. USC libraries actively support the research and scholarship of USC's faculty, students, and staff. The USC library system has 23 libraries, information centers, and the USC Digital Library.

Equipment

No major items of equipment will be used in the activities proposed in this application.

October 2019

Review committee
Transition to Aging Research Award for Predoctoral Students (F99/K00)
National Institute for Aging

Dear Sir/Madam:

RE: **KEXIN YU, PHD CANDIDATE AT THE UNIVERSITY OF SOUTHERN CALIFORNIA (USC) -
SUZANNE DWORAK-PECK SCHOOL OF SOCIAL WORK**

I am pleased to submit this letter of certification for Ms. Kexin Yu's F99/K00 (RFA-AG-20-009) proposal, under the primary sponsorship of Drs. Iris Chi, and confirm her eligibility for this award. Ms. Yu is in excellent academic standing with respect to the timeline set forth by the USC School of Social Work Ph.D. program. The F99/K00 serves as a timely opportunity for her training at the dissertation research stage and beyond.

As the Chair of the Ph.D. program at USC Suzanne Dworak-Peck School of Social Work, I certify the following:

- [REDACTED]
2. Ms. Yu has passed her qualifying exam and advanced to candidacy. She has completed all requirements for the doctoral degree except for the dissertation.
3. I certify that the USC Suzanne Dworak-Peck School of Social Work's facilities and resources will provide the necessary support for the completion of the proposed research. Ms. Yu is engaged in a multi-disciplinary scientific environment and will be mentored and supported by a highly experienced group of senior faculty members. USC provides an extensive system of library resources, computer labs, qualitative and quantitative analytic software, and IT support to Social Work Ph.D. Candidates.

University of Southern California

■ [REDACTED]

■ [REDACTED]

[REDACTED]

Obtaining the F99/K00 award will support this outstanding young scientist to receive exceptional training and build collaborative relationships with leaders with similar research interests. The transition award will facilitate Ms. Yu toward becoming an independent investigator, achieving her long-term career goals of making significant contributions to aging research and practice.

[REDACTED]

Michael Hurlburt, Ph.D.
Director of Doctoral Programs and Chair of the PhD Program
USC Suzanne Dworak-Peck School of Social Work
University of Southern California

[REDACTED]
[REDACTED]
[REDACTED]

[REDACTED]

RESEARCH & RELATED Senior/Key Person Profile (Expanded)

PROFILE - Project Director/Principal Investigator				
Prefix:	First Name*: Kexin	Middle Name	Last Name*: Yu	Suffix:
Position/Title*:	Graduate Student			
Organization Name*:	University of Southern California			
Department:	Social Work			
Division:				
Street1*:				
Street2:				
City*:				
County:				
State*:				
Province:				
Country*:				
Zip / Postal Code*:				
Phone Number*:		Fax Number:		
E-Mail*:				
Credential, e.g., agency login:				
Project Role*: PD/PI	Other Project Role Category:			
Degree Type: MSW	Degree Year: 2018			
Attach Biographical Sketch*:	File Name:	Biosketch_KY_10_ .pdf		
Attach Current & Pending Support:	File Name:			

PROFILE - Senior/Key Person			
Prefix: Professor	First Name*: Iris	Middle Name	Last Name*: Chi Suffix:
Position/Title*:	Professor		
Organization Name*:	University of Southern California		
Department:	Social Work		
Division:			
Street1*:			
Street2:			
City*:			
County:			
State*:			
Province:			
Country*:			
Zip / Postal Code*:			
Phone Number*:		Fax Number:	
E-Mail*:			
Credential, e.g., agency login:			
Project Role*: Other (Specify)	Other Project Role Category: Sponsor		
Degree Type: DSW	Degree Year: 1985		
Attach Biographical Sketch*:	File Name:	Revised_Biosketch_Ch[REDACTED].pdf	
Attach Current & Pending Support:	File Name:		

PROFILE - Senior/Key Person			
Prefix: Dr.	First Name*: Shinyi	Middle Name	Last Name*: Wu Suffix:
Position/Title*:	Associate Professor		
Organization Name*:	University of Southern California		
Department:	Social Work		
Division:			
Street1*:			
Street2:			
City*:			
County:			
State*:			
Province:			
Country*:			
Zip / Postal Code*:			
Phone Number*:		Fax Number:	
E-Mail*:			
Credential, e.g., agency login:			
Project Role*: Other (Specify)	Other Project Role Category: Co-Sponsor		
Degree Type: Ph.D.	Degree Year: 2000		
Attach Biographical Sketch*:	File Name:	Bio_Wu_10_08_19_Mentoring1016533163.pdf	
Attach Current & Pending Support:	File Name:		

PROFILE - Senior/Key Person				
Prefix: Dr.	First Name*: Lawrence	Middle Name A.	Last Name*: Palinkas	Suffix:
Position/Title*:	Professor			
Organization Name*:	University of Southern California			
Department:	Social Work			
Division:				
Street1*:	[REDACTED]			
Street2:				
City*:	[REDACTED]			
County:	[REDACTED]			
State*:	[REDACTED]			
Province:				
Country*:	[REDACTED]			
Zip / Postal Code*:	[REDACTED]			
Phone Number*:	[REDACTED]	Fax Number:	[REDACTED]	
E-Mail*:	[REDACTED]			
Credential, e.g., agency login:	[REDACTED]			
Project Role*: Other (Specify)	Other Project Role Category: Co-Sponsor			
Degree Type: Ph.D.	Degree Year: 1981			
Attach Biographical Sketch*:	File Name:	Palinkas_NIH_biosketch_for_Yu [REDACTED].pdf		
Attach Current & Pending Support:	File Name:			

PROFILE - Senior/Key Person				
Prefix:	First Name*: [REDACTED]	Middle Name [REDACTED]	Last Name*: [REDACTED]	Suffix:
Position/Title*:	[REDACTED]			
Organization Name*:	[REDACTED]			
Department:	[REDACTED]			
Division:				
Street1*:	[REDACTED]			
Street2:				
City*:	[REDACTED]			
County:	[REDACTED]			
State*:	[REDACTED]			
Province:				
Country*:	[REDACTED]			
Zip / Postal Code*:	[REDACTED]			
Phone Number*:	[REDACTED]	Fax Number:	[REDACTED]	
E-Mail*:	[REDACTED]			
Credential, e.g., agency login:	[REDACTED]			
Project Role*: Other (Specify)	Other Project Role Category: Co-Sponsor			
Degree Type: Ph.D.	Degree Year: [REDACTED]			
Attach Biographical Sketch*:	File Name:	[REDACTED]		
Attach Current & Pending Support:	File Name:			

BIOGRAPHICAL SKETCH

Provide the following information for the Senior/key personnel and other significant contributors.
Follow this format for each person. **DO NOT EXCEED FIVE PAGES.**

NAME: Kexin Yu

eRA COMMONS USER NAME (credential, e.g., agency login): [REDACTED]

POSITION TITLE: Graduate Student

EDUCATION/TRAINING *(Begin with baccalaureate or other initial professional education, such as nursing, include postdoctoral training and residency training if applicable. Add/delete rows as necessary.)*

INSTITUTION AND LOCATION	DEGREE (if applicable)	Start Date MM/YYYY	Completion Date MM/YYYY	FIELD OF STUDY
Zhejiang University, Hangzhou, China	B.S.	08/2012	06/2016	Psychology
University of Southern California, Los Angeles, CA	MSW	08/2016	05/2018	Social Work
University of Southern California, Los Angeles, CA	Ph.D.	08/2016	Present	Social Work

A. Personal Statement

My long-term research interests involve examining the potential of employing technology-enhanced socioecological approaches to reduce social isolation and delay cognitive decline in older adults. My research experience and academic training to date have laid the foundation for me to investigate the relationships between physical and social living environments, Information and Communication Technology (ICT) adoption, and Older adults' social isolation experiences.

As an undergraduate student at Zhejiang University, I worked as a research assistant for two years in the Lab of Aging and Cognition, where I contributed to building an online depression screening and intervention system designed for older adults. With a three-month research internship at the University of Alberta, I engaged in the research activities of the Translational Research for Elder Care Program and participated in preparing a systematic review manuscript on barriers and facilitators of oral health care for older adults.

My research as a graduate student has focused on understanding the risk and protective factors for social isolation and loneliness, especially regarding employing ICT to enhance social wellbeing for older adults. Using three waves of Health and Retirement Study (HRS) data, I conducted self-directed research examining the longitudinal associated between internet use and loneliness with multilevel modeling methods. Internet users were found to have a lower frequency of loneliness over time. The longitudinal association was mediated by social contact – internet users maintain their contact with family and friends better than non-users. Besides empirical inquires with survey data, I participated in conducting a systematic review on the effectiveness of social isolation interventions for Older Adults.

Working with my co-mentors, Drs. Chi & Wu, I gained hands-on research experience on employing mobile health (mHealth) tools to facilitate and empower self-management of chronic conditions among Type 2 Diabetes (T2D) patients. I led the data collection of a pilot study aimed to understand the T2D care needs of Chinese and Hispanic recent immigrants in Los Angeles. Serving as a co-investigator, I worked with Drs. Chi and Wu to organize focus group discussions with older adult residents and staff at affordable retirement communities to elicit community opinions on how environments affect health outcomes. Our work at the affordable retirement communities informed my proposed dissertation research questions. Additionally, I took the co-investigator role on a caregiver self-care study adopting a user participatory approach to co-design a mobile app with the purpose of delivering the self-care intervention more effectively. This co-design study prepares me to develop technology intervention tools to address social isolation and promote cognitive health in postdoctoral training phase and beyond.

Socially isolated older adults suffer a higher risk of cognitive decline and the progress towards Alzheimer's Disease and Related Dementias (ADRD). I aspire to investigate the mechanism through which social isolation

might influence cognitive health in my postdoctoral training. My long-term career goal is to build rigorous empirical evidence to explain how and why social wellbeing is related to cognitive health, and to develop interventions that promote cognitive health through alleviating social isolation from an ecological perspective. The proposed research and training plan outlines a series of career development activities that are carefully designed to help me to achieve the long term career goal. The NIA Transition to Aging Research Award for Predoctoral Students (F99/K00) will provide me the invaluable opportunity to receive individualized training from a strong mentor team to prepare me to become an independent investigator in the field of social isolation, cognitive health, and intervention development.

B. Positions and Honors

Professional positions:

06/13 – 08/13	Research Assistant, Human Factors & Applied Cognition Lab, North Carolina State University, Raleigh, NC
01/14 – 12/15	Research Assistant, Lab of Aging and Cognition, Zhejiang University, Hangzhou, China
02/15 – 07/15	Social Work Intern, Hangzhou Lenjoy Community & Charity Service, Hangzhou, Zhejiang Province, China
01/16 – 04/16	Research Assistant, Translating Research in Elder Care Program, University of Alberta, Edmonton, Alberta, Canada
09/16 – 05/17	Social Work Intern, ABC Adult Day Health Care Center, Los Angeles, CA, US
08/18 – 12/18	Teaching Assistant, University of Southern California, Los Angeles, CA
09/16 – present	Research Assistant, Intergenerational Mobile Technology Opportunity Program (IMTOP), University of Southern California, Los Angeles, CA
05/17 – present	Research Assistant, Intergenerational Mobile Technology Opportunity Program (IMTOP) pilot study, University of Southern California, Los Angeles, CA
02/18 – present	Co-investigator, Co-Designing a Mobile App with Chinese Immigrant Caregivers for Self-Management of Their Health, University of Southern California, Los Angeles, CA
02/19 – present	Co-investigator, Intergenerational mHealth Program for Affordable Housing Communities (ImPAHC) Feasibility Study

Academic and Professional Honors:

2013 – 2016	Morningside Culture China Honor Scholar
2014	Zhejiang Province Undergraduate Innovation and Entrepreneurship Fellowship
2014 – 2015	Excellent Undergraduate Student First-class Scholarship, Zhejiang University
2015	Kefa Cen First-class Scholarship, Zhejiang University
2016	University of Alberta Research Assistant Award
2016 – present	University of Southern California, Suzanne Dworak-Peck School of Social Work Fellowship
2017	Research Council Summer PhD Research Award
2018	American Public Health Association Nobuo Maeda International Research Award
2018	Clinical and Translational Science Institute (CTSI) pilot funding
2018	Adult Mental Health and Wellbeing (AMHW) Department Pilot Award
2018	Research Council Summer PhD Research Award
2019	Stanford Longevity Center Forming Science-Industry Research Collaboration Workshop Travel Award
2019	Research Council Summer PhD Research Award
2019	Adult Mental Health and Wellbeing (AMHW) Department Pilot Award

Memberships in Professional Societies:

2017 - present	Member, Society for Social Work Research (SSWR)
2017 - present	Member, Gerontological Society of America (GSA)
2018 - present	Member, American Public Health Association (APHA)
2019 - present	Reviewer in training, The Gerontologist
2019 - present	Reviewer in training, Journal of Gerontology: Series B

C. Contributions to Science

I. Investigating Longitudinal Predictors for Social Isolation and Loneliness. The primary outcomes of my research as a graduate student has been social isolation and loneliness among older people. I examined the longitudinal relationship between internet use on loneliness; the association was mediated with social contact. I contributed to a systematic review on the effectiveness of social isolations interventions. To bridge my interests in health

- [REDACTED]
- [REDACTED]
- [REDACTED]
- [REDACTED]

Publications

- [REDACTED]
- [REDACTED]
- Du, N., **Yu, K.**, Ye, Y., & Chen, S. (2016). Validity study of Patient Health Questionnaire-9 items for Internet screening in depression among Chinese university students. *Asia-Pacific Psychiatry*.

Publications

- Hoben, M., Clarke, A., Huynh, K. T., Kobagi, N., Kent, A., Hu, H., Pereira, R.A., Xiong, T., **Yu, K.**, Xiang, H. and Yoon, M.N., (2017). Barriers and facilitators in providing oral care to nursing home residents, from the perspective of care aides: A systematic review and meta-analysis. *International journal of nursing studies*, 73, 34-51.
- [REDACTED]

Current Research Support

[REDACTED]

BIOGRAPHICAL SKETCHProvide the following information for the Senior/key personnel and other significant contributors. **DO NOT EXCEED FIVE PAGES.**

NAME: Iris Chi, MSW, DSW

eRA COMMONS USER NAME (credential, e.g., agency login): [REDACTED]

POSITION TITLE: Professor and Golden Age Association Frances Wu Chair for the Chinese Elderly

EDUCATION/TRAINING *(Begin with baccalaureate or initial professional education, include postdoctoral and residency training.)*

INSTITUTION AND LOCATION	DEGREE (if applicable)	Completion Date MM/YYYY	FIELD OF STUDY
Chinese University of Hong Kong	BS	6/1978	Social Science
California State University, San Diego	MSW	6/1981	Social Work
University of California, Los Angeles	DSW	5/1985	Social Welfare
The Gerontological Society of America & The Center for Asian American Study, University of California, Los Angeles	Postdoctoral Fellow	12/1987	Applied Gerontology and Asian American Studies

A. Personal Statement

The overall goal of the F99/K00 application aims to examine (1) the effect of living environments on social isolation by contextualizing the role of technology and (2) the effectiveness of an Information and Communication Technology-enhanced conversational intervention on the social isolation and its effect on cognitive health among older adults. Specifically, the study plans to use mixed-methods approach which includes two longitudinal studies investigating the impact of the living environments on social isolation with 8 waves of data from the National Health & Aging Trend Study (NHATS) as well as a qualitative focus group study to explore the subjective perceptions of the environmental influence on older adults' social well-being. As a primary sponsor, I will leverage my expertise to guide this project through completion. I will provide input on research design, variables construction, data analysis, data interpretation, and manuscript writing.

I have studied health and mental health issues among older Chinese populations for 30 years. I have the expertise in ethnographic, survey research, longitudinal study, secondary data analysis, and intervention research on behavioral health studies. I have been involved in seven cross-national comparative studies in this field and most of them have pointed to the fact that responsiveness to cultural needs is very important in terms of enhancing the overall quality of life. My work has been devoted to improve the health and mental health of the low-income older Chinese immigrants to promote health equity and social justice through research, education and advocacy. In addition, my research also examines the role of interpersonal relations in moderating the impact of stress on well-being of older Asians. I am an honorary professor, fellow, consultant and advisor to more than 30 local and overseas universities and research institutes. I have a good track record of successful mentorship of mentees to become independent scholars both in the United States and in Asia. Many of them are holding prestigious academic positions and contributing to behavioral health sciences. Between 2001 and 2008, I served as mentor for the China-Rochester Suicide Research Center at the University of Rochester. I received the Awards for Excellence in Mentoring from the Mellon Academic Mentoring Support Program. Currently at University of Southern California (USC) Suzanne Dworak-Peck School of Social Work, five graduate students and four Ph.D. students are working with me on immigrant health related studies. Finally, I have established very good relationships with the community health and social service providers who work closely with Asian communities in Southern California through serving on the community based organization boards, conducting research projects and providing in-service training. I have learnt a lot from these partners about the barriers in immigrants' access to health care services in Los Angeles. We will tackle these barriers by working together to design the most appropriate interventions to help these older immigrants and their caregivers to maintaining and improving their health.

B. Positions and Honors

Positions

1980 – 1983	Executive Director, Chinese Social Service Centre, San Diego, California, USA
1987 – 1998	Assistant/Associate Professor, Department of Social Work & Social Administration, The University of Hong Kong (HKU), Hong Kong
1998 – 2003	Professor, Department of Social Work & Social Administration, HKU, Hong Kong
1998 – 2004	Director, Sau Po Center on Aging, HKU, Hong Kong
2001 – 2008	Mentor, China-Rochester Suicide Research Centre, University of Rochester
2004 –	Chinese American Golden Age Association Frances Wu Chair for the Chinese Elderly, USC Suzanne Dworak-Peck School of Social Work
2004 –	Director, China Program, USC Suzanne Dworak-Peck School of Social Work
2011 – 2014	External Research Assessor, College of Liberal Art and Social Science, City University of Hong Kong, Hong Kong and Department of Social Work, National Taiwan University, Taiwan
2010 -	Senior Scientist, Global and Community Health Program Lead, USC Edward R. Roybal Institute on Aging
2016 - 2018	External Examiner, Master of Gerontology Program, Singapore University of Social Sciences (SUSS), Singapore

Honors

1999 –	Honorary Professor and Associated Health Researcher, Center on Aging, University of Victoria, British Columbia, Canada
2000 –	Fellow, interRAI
2003 –	International Consultant, HKU Jockey Club Centre for Suicide Research and Prevention
2003	Award for Research, International Psychogeriatric Association
2004	Bronze Bauhinia Star, the Government of Hong Kong Special Administrative Region, China
2005 –	Honorary Professor of Gerontology, USC Leonard Davis School of Gerontology
2006 –	Advisor & Research Fellow, Center of Research on Disability and Development, China
2006	Awards for Excellence in Mentoring, Mellon Academic Mentoring Support Program
2009	Distinguished Faculty, S.C. Franklin Award, University of Southern California
2010 -	Fellow, Gerontological Society of America (GSA)
2010	Gary Andrews Fellow, Australian Association of Gerontology, Australia
2011 –	Honorary Advisor, Sau Po Centre on Aging, The University of Hong Kong, Hong Kong
2012 –	International Advisor, Center for Aging and Health Studies, School of Public Policy and Administration, Xi'an Jiaotong University, China
2012 – 2017	Honorary Professor, Nanjing University, Nanjing, China
2013 – 2018	Fellow, Risk Society and Policy Research Center, College of Social Sciences, National Taiwan University, Taiwan
2014 –	Fellow, American Academy of Social Welfare and Social Work
2015 – 2018	Expert and Resource Person, Asian Development Bank (ADB)
2016 – 2018	Distinguished Professor, Department of Sociology, Shanghai Lixin Accounting and Finance College, Shanghai, China
2017 –	Honorary Professor, Master of Social Work Education Center, Beijing Normal University, Beijing, China
2017 – 2022	Honorary Consultant, Evidence-based Social Work Research Center, Jiangsu Civic Affairs Bureau, Nanjing, China
2018	Nabuo Maeda International Research Award, The American Public Health Association
2019	Congressional Women of the Year Award, 27th Congressional District

Journal Reviewer: (over 30 ad hoc reviewer for journals)

Editorial Board and guest editor: (16 journals) JGMS, International Journal of Social Welfare, Geriatrics and Gerontology International, Australasian Journal on Ageing, Journal of Social Work Practice and Evaluation

Associate Editor: (2 journals) BMC Geriatrics & International Journal of Social Welfare

C. Contribution to Science

1. A major part of my publications addressed health and mental health issues in older adult populations. I have been collecting longitudinal clinical data since 2002 and I lead a multidisciplinary research team to analyze the data. These publications identified the health care needs of older adults in various settings, ranging from community setting (living in their own homes) to long-term care facilities, and from policy to direct practice including management of health and long term care services. By providing evidence of these health care needs and clinical practices, healthcare providers can serve older adults in a more effective way. Below are the selected publications related to the studies of Chinese Immigrants:
 - a. Liu, J., Guo, M. & Chi, I. (eds.) (2019). Special issue on Intergenerational relationships and well-being of East Asian older adults in migrant families. *Journal of Ethnic & Cultural Diversity in Social Work*. Routledge: Taylor & Francis Group.
 - b. Tang, F., Chi, I., & Dong, X. (2019). Gender differences in the prevalence and incidence of cognitive impairment: Does immigration matter? *Journal of the American Geriatrics Society*, 67, S513–S518.
 - c. Yuan, L., Liu, L., Leu, R. & Chi, I. (2017). Design a patient empowerment program for Chinese immigrants. *Health and Social Work*, 42 (2), 121-124. doi: 10.1093/hsw/hlx005
 - d. Li, J., Xu, L., & Chi, I. (2017). Challenges and resilience of aging in the United States of older Chinese immigrants. *Aging and Mental Health*. doi:10.1080/13607863.2017.1377686
2. Another contribution area is on the studies of Chinese Caregivers:
 - a. [REDACTED]
 - b. Li, M., Mao, W., Chi, I., & Lou, V. W. (2019). Geographical proximity and depressive symptoms among adult child caregivers: social support as a moderator. *Aging & Mental Health*, 23(2), 205-213.
 - c. Chong, M. L. A., Kwan, C. W., Lou, V. W. Q., & Chi, I. (2017). Can domestic helpers moderate distress of offspring caregivers of cognitively impaired older adults? *Aging and Mental Health*, 27(10), 1023-1030. doi: 10.1080/13607863.2016.1191059
 - d. Fu, F., Zhao, H., Tong, F., & Chi, I. (2017). A systematic review of psychosocial interventions to cancer caregivers. *Frontiers in Psychology*, 8, Article 834. doi: 10.3389/fpsyg.2017.0083
3. Another contribution I made to the body of work in gerontology was exploring factors associated with Cognitive function:
 - a. Tang, F., Chi, I., & Dong, X. (2019). Gender differences in the prevalence and incidence of cognitive impairment: Does immigration matter? *Journal of the American Geriatrics Society*, 67, S513–S518.
 - b. Tang, F., Chi, I., Zhang, W. & Dong, X. (2018). Activity engagement and cognitive function: Findings from a community-dwelling U.S. Chinese Aging Population Study. *Gerontology and Geriatric Medicine*, 4, doi: 2333721418778180
 - c. Lee, Y., Chi, I. & Palinkas, L. (2018). Retirement, leisure activity engagement, and cognition. *Journal of Aging and Health*. doi: 0898264318767030
 - d. Lee, Y. & Chi, I. (2016). Do cognitive leisure activities really matter in the relationship between education and cognition? Evidence from the Aging, Demographics, and Memory Study (ADAMS). *Aging and Mental Health*, 20, 252–261.
4. Beneath are the selected publications related to health and long term care in aging population including scale construction and validation:
 - a. [REDACTED]
 - b. Lam, A., Kwan, J., Kwan, C.W. & Chi, I. (2017). Factors associated with the trend of physical and chemical restraint use amongst long-term care facility residents in Hong Kong: Data from an 11-year observational study. *Journal of the American Medical Directors Association*, 18(12), 1043-1048.
 - c. Luo, H., Lum, T.Y., Wong, G.H., Kwan, J.S., Tang, J.Y., & Chi, I. (2015). Predicting adverse health outcomes in nursing homes: A 9-year longitudinal study and development of the FRAIL-Minimum Data Set (MDS) Quick Screening Tool. *Journal of the American Medical Directors Association*, 16(12), 1042-1047.
 - d. Li, H., Xu, L., & Chi, I. (2015). Factors related to Chinese older adults' suicidal thoughts and attempts. *Aging and Mental Health*. doi:10.1080/13607863.2015.1037242

<https://www.ncbi.nlm.nih.gov/sites/myncbi/iris.chi.1/bibliography/51376360/public/?sort=date&direction=ascending>

Ongoing Research Support

[illegible]

BIOGRAPHICAL SKETCH

Provide the following information for the Senior/key personnel and other significant contributors.
Follow this format for each person. **DO NOT EXCEED FIVE PAGES.**

NAME: Shinyi Wu

eRA COMMONS USER NAME (credential, e.g., agency login): [REDACTED]

POSITION TITLE: Associate Professor of Social Work and Industrial and Systems Engineering

EDUCATION/TRAINING *(Begin with baccalaureate or other initial professional education, such as nursing, include postdoctoral training and residency training if applicable. Add/delete rows as necessary.)*

INSTITUTION AND LOCATION	DEGREE (if applicable)	Completion Date MM/YYYY	FIELD OF STUDY
Chung-Yuan Christian University, Taiwan	B.S.	06/1992	Industrial Engineering
University of Wisconsin – Madison	M.S.	12/1993	Industrial Engineering
University of Wisconsin – Madison	Ph.D.	05/2000	Industrial Engineering

A. Personal Statement

I will serve as a co-sponsor for Kexin Yu's proposed F99/K00 application titled "Examining the Longitudinal Influence of the Physical and Social Environments on Social Isolation and Cognitive Health: contextualizing the role of technology". I have the research expertise and mentoring experiences relevant to the proposed study to support Kexin in successfully carrying out the aims of the study. I have a multidisciplinary background in industrial and systems engineering (ISE), health services and policy research, health economics, and social work. My research has focused on enhancing quality of health systems and reducing health disparities through innovative design and implementation of information and digital health technology, cost-effectiveness analysis and economic evaluation, application of process and quality improvement methods, and decision analysis and simulation. I have successfully led a large-scale study called Diabetes-Depression Care-Management Adoption Trial (DCAT) for low-income, midlife, primarily Latino diabetic populations in 8 safety-net clinics (1406 patients). The DCAT tested a home-grown telehealth technological system integrating automated assessment of patient symptoms and health behaviors monitoring with disease management registry to implement evidence-based comorbid depression care management. The trial results showed the intervention's effects on improving health outcomes and reducing costs. I have also led digital health studies to design and test a text-messaging program for exercise prompting for low-income, limited English proficiency Latino diabetes patients, and a mHealth technology for team-based care management to support stroke prevention. In addition, I have led a trial testing an intergenerational self-management program with tablet technology and a mobile app to improve health outcomes among older adults with diabetes. Recently, I developed and pilot-tested a mobile app for Chinese caregivers with limited English proficiency to provide them self-management knowledge and behavioral support for reducing caregiving burdens and better health outcomes. I have trained Kexin in the latter two projects aforementioned and advised her on her own doctoral research aimed to disentangle the directionality and time sequence between the relationships of geriatric conditions and loneliness among older adults. I believe my previous experiences and expertise in technology, health services research, and in leading trans-disciplinary, digital health-related clinical trials to improve health outcomes for older adults makes me well qualified to serve as the co-sponsor and nicely complements the expertise of the rest of the sponsor team.

- Wu, S.**, Vidyanti, I., Liu, P., Hawkins, C., Ramirez, M., Guterman, J., Gross-Schulman, S., Sklaroff, L., & Ell, K. (2014). Patient-centered technological assessment and monitoring of depression for low-income patients. *Journal of Ambulatory Care Management*, 37(2), 138–147. PMID: 24525531. doi:10.1097/JAC.0000000000000027.
- Wu, S.**, Ell, K., Jin, H., Vidyanti, I., Chou, C-P., Lee, P-J., Gross-Schulman, S., Sklaroff, L., Belson, D., Nezu, A.M., Hay, J., Wang, C-J., Scheib, G., Di Capua, P., Hawkins, C., Liu, P., Ramirez, M., Wu, B.W., Richman, M., Myers, C., Gonzalez, J., Agustines, D., Dasher, R., Kopelowicz, A., Allevato, J.,

- Roybal, M., Ipp, E., Haider, U., Lakshaman, R., Graham, S. Mahabadi, V., & Guterman, J. (2018). Comparative effectiveness of a technology-facilitated depression care management model in safety-net primary care patients with type 2 diabetes: 6-month outcomes of a large clinical trial. *Journal of Medical Internet Research*, 20(4), e147. PMID: PMC5938593
- c. Ramirez, M., **Wu, S.**, Ryan, G., Towfighi, A., Vickrey, B. (2017). Using Beta-version mHealth technology for team-based care management to support stroke prevention: An assessment of utility and challenges. *JMIR Research Protocols*, 6(5), e94. PMID: PMC5461415
- d. **Wu, S.**, Ell, K., Jin, H., ...& Guterman, J. (2018). Comparative effectiveness of a technology-facilitated depression care management model in safety-net primary care patients with type 2 diabetes: 6-month outcomes of a large clinical trial. *Journal of Medical Internet Research*, 20(4), e147. PMID: 29685872, doi: 10.2196/jmir.7692.

B. Positions and Honors

Positions and Employment

1999-2006	Associate Engineer, Health Program, RAND, Santa Monica, CA
2006-2007	Engineer, Health Program, RAND, Santa Monica, CA
2006-2009	Associate Director, Roybal Center for Health Policy Simulation, RAND
2008-2014	Assistant Professor, Daniel Epstein Department of Industrial and Systems Engineering, University of Southern California, Los Angeles, CA
2008-2012	Affiliated Adjunct Engineer, Technology & Applied Science Group and Health Program, RAND
2014-present	Associate Professor, School of Social Work and Daniel J. Epstein Department of Industrial and Systems Engineering, University of Southern California, Los Angeles, CA
2015-2016	Associate Director of Social and Health Services Research, Edward R. Roybal Institute on Aging, University of Southern California, Los Angeles, CA
2016-2017	Associate Director, Center for Artificial Intelligence in Society, University of Southern California, Los Angeles, CA

Other Experience

2008-2011	National Science Foundation (NSF) Peer Review Committee: Service Enterprise Systems
2011-2014	Committee Member, Quality Committee of the Hospital Governing Boards of USC University Hospital and USC Norris Cancer Hospital
2012-2015	Technical Consultant to US Department of Veterans Affairs Health Services Research & Development: Center of Excellence (COE) and Center for the Study of Healthcare Innovation, Implementation & Policy (CSHIIP)
2013	Expert Testimony at Informational Hearing on Improving Outcomes through the Patient Centered Medical Home, Assembly Committee on Health, Assembly California Legislature.
2013-2014	Primary Care Practice Facilitation Technical Expert Panel to the Agency for Healthcare Research and Quality (AHRQ)

Honors

2004	Most Outstanding Abstract, AcademyHealth 21 th Annual Research Meeting, San Diego, CA
2007	RAND Merit Bonus Award for Improve Policy and Decisionmaking through Research and Analysis, Santa Monica, CA
2007	Honor of Service for Quality, Chinese Society for Quality, Taipei, Taiwan
2008	Faculty Development Award, Chinese American Faculty Association of Southern California, Los Angeles, CA
2014	Best Poster Paper Award, International Conference on Big Data and Analytics in Health Care
2014	Honorable Mention, Preventing Chronic Disease Annual Student Paper Contest, National Center for Chronic Disease Prevention and Health Promotion, U.S. Centers for Disease Control and Prevention (With Doctoral student Brian Wu et al.)
2015	Outstanding Teacher of the Year, Daniel J. Epstein Department of Industrial and Systems Engineering, University of Southern California
2015	Sterling C. Franklin Award for Distinguished Faculty, School of Social Work, University of Southern California
2015	Honorable Mention, Preventing Chronic Disease Annual Student Paper Contest, National Center for Chronic Disease Prevention and Health Promotion, U.S. Centers for Disease Control and Prevention (With Doctoral student Haomiao Jin et al.)

- 2015 Top Rated Abstract, Society for Medical Decision Making 37th Annual North American Meeting, St. Louis, MO
- 2018 Award for Excellence in Mentoring in the category of Faculty Mentoring Graduate Students, University of Southern California
- 2018 Nobu Maeda International Research Award, American Public Health Association

C. Contributions to Science

1. My research contributes to implementation science to reduce health disparities in safety net chronic illness patients. In the context of a national evaluation of innovative language access services for Spanish-speaking patients with limited English proficiency, I led the analysis of implementation costs and lessons learned. The key lessons learned are that there are a variety of creative models that can be employed to develop, enhance, and sustain language access services, but there are considerable operating costs and many internal and external system changes required for their successful implementation and sustainable adoption. More recently, in partnership with the LA County Department of Health Services (DHS), I led a team to design the DCAT cutting-edge automated telephonic assessment system and integrated the telecommunications with the clinical registry system to prompt providers with tasks and alerts. This solution automates individualized, regular assessment of a patient's condition (based on patient medical history and preferences such as language), thus identifying their unmet needs in a timelier manner and strengthening emotional support. To test this technology, I conducted a quasi-experimental 3-group trial with 1,406 low-income, predominantly Hispanic/Latino diabetes patients served by DHS safety-net clinics. Sample DCAT publications include:

- a. **Wu, S.**, Ridgely, M. S., Escarce, J. J., & Morales, L. S. (2007). Language access services for Latinos with limited English proficiency: Lessons learned from *Hablamos Juntos*. *Journal of General Internal Medicine*, 22(Suppl 2), 350–355. PMCID: PMC2078542
- b. **Wu, S.**, Ell, K., Gross-Schulman, S. G., Sklaroff, L. M., Katon, W. J., Nezu, A. M., Lee, P.-J., Vidyanti, I., Chou, C.-P., & Guterman, J. J. (2014). Technology-facilitated depression care management among predominantly Latino diabetes patients within a public safety net care system: Comparative effectiveness trial design. *Contemporary Clinical Trials*, 37(2), 342–354. PMID: 24215775
- c. **Wu, S.**, Duan, N., Wisdom, J. P., Kravita, R. L., Owen, R. R., Sullivan, G., Wu, A.W., Di Capua, P., & Hoagwood, K. E. (2014). Integrating science and engineering to implement evidence-based practices in health care settings. *Administration and Policy in Mental Health and Mental Health Services Research*, 42(5), 588-592. PMCID: PMC4363001
- d. **Wu, S.**, Ell, K., Jin, H., Vidyanti, I., Chou, C-P., Lee, P-J., Gross-Schulman, S., Sklaroff, L., Belson, D., Nezu, A.M., Hay, J., Wang, C-J., Scheib, G., Di Capua, P., Hawkins, C., Liu, P., Ramirez, M., Wu, B.W., Richman, M., Myers, C., Gonzalez, J., Agustines, D., Dasher, R., Kopelowicz, A., Allevato, J., Roybal, M., Ipp, E., Haider, U., Lakshaman, R., Graham, S. Mahabadi, V., & Guterman, J. (2018). Comparative effectiveness of a technology-facilitated depression care management model in safety-net primary care patients with type 2 diabetes: 6-month outcomes of a large clinical trial. *Journal of Medical Internet Research*, 20(4), e147. PMCID: PMC5938593

2. My cost-effectiveness analysis work aims to facilitate implementation of evidence-based health interventions and programs. In many preventive and health service areas, cost-effectiveness analyses can provide critical information to help policy-makers and community planning organizations devise strategies that maximize the use of scarce resources to promote population health and social justice. To advance that goal, I have conducted cost-effectiveness analysis and analytic modeling in many health-related areas, including HIV prevention, falls prevention, physical activity promotions, and technology-facilitated care-management.

- a. **Wu, S.**, Keeler, E. B., Rubenstein, L., Maglione, M., & Shekelle, P. (2010). A cost-effectiveness analysis of a proposed national falls prevention program. *Clinics in Geriatric Medicine*, 26(4), 751-766. PMID: 20934620
- b. **Wu, S.**, Shi, Y., Cohen, D. A., Pearson, M. L., & Sturm, R. (2011). Economic analysis of physical activity interventions. *American Journal of Preventive Medicine*, 40(2), 149–158. PMCID: PMC3085087
- c. Babey, S., **Wu, S.**, & Cohen, D. A. (2014). How can schools help youth increase their physical activity? An Economic analysis comparing school-based programs. *Preventive Medicine*, 69, S55–S60. PMID: 25456799

- a. Vidyanti, I., Wu, B., & **Wu, S.** (2015). Low-income minority patient engagement with automated telephonic depression assessment and impact on health outcomes. *Quality of Life Research*, 24(5), 1119-1129, PMID: PMC4412647
- b. Di Capua, P., Wu, B., Sednew, R., Ryan, G., & **Wu, S.** (2016). Complexity in redesigning depression care: Comparing intention vs. implementation of an automated depression screening and monitoring program. *Population Health Management*, 19(5), 349-56. PMID: 27028043
- c. Ramirez, M. & **Wu, S.** (2017). Phone messaging to prompt physical activity and social support among low-income Latino patients with type 2 diabetes: A randomized pilot study. *JMIR Diabetes* 2(1), e8. PMID: 30291094
- d. Jin, H. & **Wu, S.** (2019). The validity of conducting depression screening via text messaging vs. interview assessment: Randomized controlled trial. *JMIR Research Protocols* 2019 8(3): e12392. PMID: 30924787

- a. Coleman, C., Pearson, M., & **Wu, S.** (2011). Integrating chronic care and business strategies in the safety net: A practice coaching manual. In R. Simons (Ed.), *Operations Management: A Modern Approach* (pp. 267–296). Canada: Apple Academic Press, Inc., 2011. ISBN: 978-1-926692-90-6.
- b. **Wu, S.**, Liu, P., & Belson, D. (2013). Multiple-hospital lean initiative to improve congestive heart failure care: A mixed-methods evaluation. *Journal of the Society of Healthcare Improvement Professionals*, 3.
- c. Jin, H., **Wu, S.**, & Di Capua, P. (2015). Development of a clinical forecasting model to predict comorbid depression among diabetes patients and an application in depression screening policy making. *Preventing Chronic Disease*, 12, E142. PMID: PMC4561536 (Selected as Honorable Mention in the 2015 Preventing Chronic Disease Annual Student Paper Contest.)
- d. Alibraham, A., & **Wu, S.** (2018). An agent-based simulation model of patient choice of health care providers in accountable care organizations. *Health Care Management Science*, 21(1), 131-143. PMID: 27704322

<http://www.ncbi.nlm.nih.gov/sites/myncbi/shinyi.wu.1/bibliography/50282292/public/?sort=date&direction=ascending>

[REDACTED]

[REDACTED]

Completed Research Support

[REDACTED] 07/01/2014 – 08/31/2019

[REDACTED]

CBET-1548502

Mataric (PI)

09/01/2015 – 08/31/2019

National Science Foundation

EAGER: Studying the Dynamics of In Home Adoption of Socially Assistive Robot Companions for Elderly

Emotional and social factors play a key role in staying well longer and on improved health outcomes across the age span. There is a clear need for ethical, age-appropriate, and personalized companion technologies for the elderly and their caregivers, and studies already show a user preference for embodied (robot) companions over screen-based ones. This project will iteratively develop and probe the effects of introducing an in-home socially assistive robot at multiple levels (with the older adult user, with an in-home family caregiver, and with other family members) and in different interaction contexts (user-caregiver and user-visitors).

Role: Co-Principal Investigator

[REDACTED]

U54NS081764

Vickrey (PI)

09/01/2012 – 08/31/2017

National Institute of Neurological Disorders and Stroke, NIH

Los Angeles Stroke Prevention/Intervention Research Program in Health Disparities

This program is a multi-partner research center that will conduct two randomized, controlled, community-based trials of stroke-prevention interventions. My research is to design a mobile health technology application and an evaluation to measure perceptions of usefulness, ease of use, and willingness to adopt the mobile health technology from multiple stakeholders' perspectives. This research will support one of the center's aims, which is to create and test sustainable interventions to redress racial/ethnic disparities in the occurrence of stroke.

Role: Subaward Co-Principal investigator

[REDACTED]

BIOGRAPHICAL SKETCH

Provide the following information for the Senior/key personnel and other significant contributors.
Follow this format for each person. **DO NOT EXCEED FIVE PAGES.**

NAME: Lawrence A. Palinkas

eRA COMMONS USER NAME (credential, e.g., agency login): [REDACTED]

POSITION TITLE: Albert G. and Frances Lomas Feldman Professor of Social Policy and Health

EDUCATION/TRAINING (*Begin with baccalaureate or other initial professional education, such as nursing, include postdoctoral training and residency training if applicable. Add/delete rows as necessary.*)

INSTITUTION AND LOCATION	DEGREE (if applicable)	Completion Date MM/YYYY	FIELD OF STUDY
University of Chicago, IL	B.A.	06/1974	Anthropology
University of California, San Diego	M.A.	06/1975	Anthropology
University of California, San Diego	Ph.D.	06/1981	Anthropology

1. Personal Statement

My extensive content expertise in older adult mental health and implementation science and my methodological expertise in mixed method designs position me as an ideal participant in the proposed project. In my own research program, I am particularly interested in the role of social networks and the use of research evidence in the implementation of evidence-based practices for prevention and treatment of behavioral and mental health disorders. I have held positions of leadership in studies that expanded my understanding of the several aspects of the current proposal. I conducted research on the implementation of a psychoeducational intervention designed to prevent HIV and STD incidence in female sex workers in Mexico. I am the Principal Investigator of a NIDA-funded study designed to develop a system for measuring sustainment of SAMHSA-funded substance use and suicide prevention programs. I have been engaged in developing new types of mixed method designs that target implementation of evidence-based practices and addressing the behavioral health needs of older adults. Finally, my own interest and expertise in the areas discussed above gives me a profound appreciation of the research and community impact of the proposed grant, such that I am deeply committed to its execution and successful completion.

1. **Palinkas LA**, Aarons GA, Horwitz SM, Chamberlain P, Hurlburt, M, Landsverk J. Mixed method designs in implementation research. *Admin Policy Ment Health*; 2011; 38:44-53. PMID 20967495
2. **Palinkas LA**, Horwitz SM, Green CA, Wisdom JP, Duan N, Hoagwood KE. Purposeful sampling for qualitative data collection and analysis in mixed method implementation research. *Admin Policy Ment Health*, 2015; 42:533-544. doi: 10.1007/s10488-013-0528-y. PMID:24193818
3. **Palinkas LA**, Spear SE, Mendon SJ, Villamar J, Valente T, Chou CP, Landsverk J, Kellam SG, Brown CH. Measuring sustainment of prevention programs and initiatives: A study protocol. *Implement Sci*, 2016; 11:95 doi: 10.1186/s13012-016-0467-6 PMID:27422149
4. **Palinkas LA**, Zatzick D. Rapid assessment procedure informed clinical ethnography (RAPICE) in Pragmatic clinical trials of mental health services implementation: Methods and applied case study. *Admin Policy Ment Health*; 2018, <https://doi.org/10.1007/s10488-018-0909-3> PMID:30488143

B. Positions and Honors

1981-1982	Statistician, Naval Health Research Center (NHRC), San Diego, CA
1981-1984	Visiting Lecturer, University of California, San Diego (UCSD)
1982-1984	National Research Council Postdoctoral Research Associate, NHRC, San Diego, CA
1984-1989	Research Psychologist, Naval Health Research Center, San Diego, CA
1986-1989	Assistant to Associate Adjunct Professor, Department of Community and Family Medicine, UCSD
1993-2005	Associate to Professor, Department of Family and Preventive Medicine, UCSD

1996-2005	Director, Immigrant/Refugee Health Studies, UCSD
1999-2005	Professor, UCSD/San Diego State University Joint Doctoral Program in Clinical Psychology
1998-2005	Professor, UCSD/San Diego State University Joint Doctoral Program in Public Health
2005-2014	Adjunct Professor, Dept of Family and Preventive Medicine, UCSD
2005-present	Professor, School of Social Work and Department of Anthropology, University of Southern California (USC), Los Angeles
2007-present	Professor, Department of Preventive Medicine, USC
2008-present	Albert G and Frances Lomas Feldman Professor of Social Policy and Health, USC
2009-2014	Adjunct Professor, Department of Medicine, UCSD
2014-2019	Chair, Department of Children, Youth and Families, School of Social Work, USC

Other Experience and Professional Memberships

1992	Elected Fellow, American Anthropological Association
1996-present	United States Representative to the Human Biology and Medicine Working Group, Scientific Committee on Antarctic Research
1997-2000	Member, Committee on Space Biology and Medicine, Space Studies Board, National Research Council/ National Academy of Sciences
2003-2005	Chair, External Advisory Council, National Space Biomedical Research Institute
2004-2005	Member, Committee on Reviewing NASA's Biomedical Critical Path Roadmap, Institute of Medicine
2006-2010	Member, Services Review Non-Specialty IRG, National Institute of Mental Health
2009-2010	Member, Decadal Survey on Physical and Biological Sciences in Space: Human Behavior and Mental Health Panel, Space Studies Board, National Research Council/National Academy of Sciences
2010-11	Member, Committee to Review the Federal Response to the Health Effects Associated with the Gulf of Mexico Oil Spill, Institute of Medicine.
2013	Member, Committee on Ethics Principles and Guidelines for Health Standards for Long Duration and Exploration Spaceflights, Institute of Medicine
2013-16	Member, Forum on Child Cognitive, Affective and Behavioral Development, Institute of Medicine
2014-	Member, Committee on Aerospace Medicine and Medicine of Extreme Environments, Institute of Medicine

Honors

1989	Antarctic Service Medal (National Science Foundation/U.S. Navy)
1998	Visiting Professor, Peking Union Medical College, Beijing, China
2002	Deputy Chief Officer, Life Sciences Standing Scientific Committee, Scientific Committee on Antarctic Research
2004-2006	Visiting Professor, University of Oulu, Finland
2010	Sterling C. Franklin Award for Outstanding School of Social Work Faculty, USC
2011-12	Visiting Professor, School of Public Health, University of California, Berkeley
2015	Excellence in Leadership and Creativity Award, School of Social Work, USC
2015	Elected Fellow, American Academy of Social Welfare and Social Work
2017	Elected Fellow, Society for Social Work and Research

C. Contributions to Science

1. I have devoted much of my career to the challenges of developing innovative approaches to integrating quantitative and qualitative methods in mixed methods designs for health services research. This work has led to development of guidelines for using qualitative and mixed methods in NIH and PCORI-funded research and for sampling study participants in mixed method implementation research. It has also led to the creation of new methods for measuring processes and outcomes of implementation and sustainment of evidence-based programs and practices, including a rapid assessment version of clinical ethnography, procedures for quantifying qualitative data, use of social network analysis to develop and tailor strategies to facilitate implementation, and assessment of sustainment of prevention programs and initiatives.
 - a. Valente T, **Palinkas LA**, Czaja S, Chu KH, Brown CH. Social network analysis for program implementation (SNAPI). *PLOS One*, 2015; 10(6):e0131712. doi: 10.1371/journal.pone.0131712 PMID:26110842

- b. Zatzick DF, Russo J, Darnell D, Chambers DA, **Palinkas LA**, Van Eaton E, et al. An effectiveness-implementation hybrid trial study protocol targeting posttraumatic stress disorder and comorbidity. *Implement Sci*, 2016; 11(1):58. doi: 10.1186/s13012-016-0424-4. PMID:27130272
 - c. Brown CH, Curran G, **Palinkas LA**, Wells KB, Jones L, Collins LM, et al. An overview of research and evaluation designs for dissemination and implementation. *Ann Rev Public Health*, 2017; 38:1-22. doi:10.1146/annurev-publhealth-021816-044215. PMID:28384085
 - d. **Palinkas LA**, Mendon SJ, Hamilton AB. Innovations in mixed methods evaluations. *Ann Rev Public Health*, 2019; 40:20.1–20.20. <https://doi.org/10.1146/annurev-publhealth-040218-044215>
 2. My current work is focused on the challenges involved in overcoming the wide gap in research to practice through understanding barriers to the implementation of evidence-based practices in mental health services and disease prevention, and developing and evaluating strategies designed to overcome these barriers. My contribution to this work has been in understanding the influence of social networks and use of research evidence in evidence-based practice implementation and in the application of novel mixed method designs in implementation research. This work has demonstrated that EBP implementation requires effective partnerships and collaborations that exchange information and resources through influence networks governed by the sharing of distributed understandings.
 - a. Aarons GA, **Palinkas LA**. Implementation of evidence-based practice in child welfare: service provider perspectives. *Admin Policy Ment Health*, 2007; 34(4):411-419. PMID:17410420
 - b. **Palinkas LA**, Holloway IW, Rice E, Fuentes D, Wu Q, Chamberlain, P. Social networks and implementation of evidence-based practices in public youth-serving systems: a mixed methods study. *Implementation Sci*, 2011; 6:113. doi: 10.1186/1748-5908-6-113. PMID:21958674
 - c. **Palinkas LA**, Um MY, Jeong CH, Chor KH, Olin S, Horwitz SM, Hoagwood KE. Adoption of innovative and evidence-based practices for children and adolescents in state-supportive mental health clinics: a qualitative study. *Health Res Policy Syst*, 2017; 29;15(1):27. doi: 10.1186/s12961-017-0190-z. PMID:28356145
 - d. **Palinkas LA**. *Achieving implementation and exchange: The science of delivering evidence-based practices to at-risk youth*. Bristol, UK: Policy Press, 2019.
 3. One of the key elements of successful implementation that has been the focus of my work has been the use of research evidence in health care policy and practice. I was perhaps the first to develop and evaluate an instrument designed to measure the use of research evidence. My research also found that such use was independently associated with the implementation of evidence-based policies and practices and child welfare and child mental health.
 - a. Hyde J, Mackie TI, **Palinkas LA**, Niemi E, Leslie LK. Evidence use in mental health policy making for children in foster care. *Admin Policy Ment Health* 2016; 43(1):52-66. doi: 10.1007/s10488-015-0633-1. PMID:25711392
 - b. Garcia AR, Kim M, **Palinkas LA**, Snowden L, Landsverk J. Socio-contextual determinants of research evidence use in public-youth systems of care. *Admin Policy Ment Health*, 2016; 43(4):569-78. doi: 10.1007/s10488-015-0640-2. PMID:25702145
 - c. **Palinkas LA**, Garcia AR, Aarons GA, Finno M, Holloway IW, Mackie T, et al. Measuring use of research evidence in child-serving systems: the Structured Interview for Evidence Use (SIEU). *Res Soc Work Pract*, 2016; 26(5):550-564. PMID:27616869
 - d. **Palinkas, LA**, Saldana L, Chou CP, Chamberlain, P. Use of research evidence and implementation of evidence-based practices in youth-serving systems. *Child Youth Serv Rev*, 2017; 83:242-247. PMID:29170572
 4. I have also devoted considerable effort in understanding and preventing physical and mental health disorders in older adults. I examined associations between depressive symptoms in cardiovascular disease, obesity, diabetes, and social networks in older adults. Along with my student, Ross Browner, I was one of the first investigators to document an association between depressive symptoms and low cholesterol in older men. I also found evidence to support the Jolly Fat hypothesis that posits an inverse association between body mass and depressive symptoms, in men but not women.
 - a. **Palinkas LA**, Wingard DL, Barrett-Connor E. Chronic illness and depressive symptoms in the elderly: a population-based study. *J Clin Epidemiol*, 1990; 43:1131-1141. PMID: 2243251

- b. Morgan RE, **Palinkas LA**, Barrett-Connor E, Wingard DL. Plasma cholesterol and depressive symptoms in older men: a population-based study. *Lancet*, 1993; 341:75-79. PMID:8093404
 - c. Barrett-Connor E, **Palinkas LA**. Low blood pressure and depression in older men: a population-based study. *BMJ*, 1994; 308:446-449. PMID:8124175
 - d. **Palinkas LA**, Wingard DL, Barrett-Connor E. Depressive symptoms in overweight and obese older adults: A test of the "jolly fat" hypothesis. *J Psychosom Res*, 1996; 40(1):59-66. PMID: 8730645
5. I conducted qualitative studies of the needs for mental health services and the common elements of successful aging in older adults. The former set of studies found that age-appropriate and culturally appropriate solutions that address both prevention and treatment may represent the best strategies for addressing the challenges of mental illness. The more recent work has identified core personal characteristics and environmental factors that contribute to successful aging and the modulation between actively accepting end of life while continuing the desire to grow and change at this critical time.
- a. **Palinkas LA**, Criado V, Fuentes D, Shepherd S, Milian H, Folsom D, and Jeste DV. Unmet needs for services for older adults with mental illness: Comparison of views of different stakeholder groups. *Am J Geriatr Psychiatry*, 2007; 15:530-540. PMID:17545452
 - b. Reichstadt J, Depp CA, **Palinkas LA**, Folsom DP, Jeste DV. Building blocks of successful aging: A focus group study of older adults' perceived contributors to successful aging. *Am J Geriatr Psychiatry*, 2007; 15:194-201. PMID:17322132
 - c. Reichstadt J, Sengupta G, Depp CA, **Palinkas LA**, Jeste DJ. Older adults' perspective on successful aging: qualitative interviews. *Am J Geriatr Psychiatry*, 2010; 18:567-575. PMID:20593536
 - d. Montross-Thomas LP, Joseph J, Edmonds EC, **Palinkas LA**, Jeste DV. Reflections on wisdom at the end of life: qualitative interviews of 21 hospice patients. *Int Psychogeriatrics*, 2018; 30(12): 1759-1766. PMID:29361991

List of Published Work in my Bibliography (from over 270 peer-reviewed publications):

<http://www.ncbi.nlm.nih.gov/sites/myncbi/lawrence.palinkas.1/bibliography/47624041/public/?sort=date&direction=ascending>

D. Additional Information: Research Support and/or Scholastic Performance

Ongoing Research Support

Northwestern University (Sub), 60044679USC Brown (PI) 9/1/16 – 6/30/21
NIH-NIDA P30 DA027828

Center for Prevention Implementation Methods for Drug Abuse and Sex Risk Behavior
To integrate and extend system science methods to address critical research challenges of federally funded implementation research aimed at preventing drug use and sexual risk behaviors.
Role: Site Principal Investigator

New York University (Sub) Hoagwood (PI) 9/21/19 – 4/30/23
NIH-NIMH P50 MH113662-01A1

Accelerator Strategies for States to Improve System Transformations Affecting Children, Youth and Families
To develop tools to improve the uptake of child mental health (MH) research evidence by state policymakers, and test two practical accelerator strategies to improve D&I of evidence-based practices in state systems.
Role: Site Principal Investigator and Co-PI, Methods Core

University of Maryland Baltimore (Sub), F211896-2 Dubowitz (PI) 4/1/19 – 8/31/23
NIH-NICHD R01 HD092489-01A1

Dissemination and implementation of the Safe Environment for Every Kid (SEEK) Model for Preventing Child Maltreatment.
To evaluate the impact of independent online (IND) versus in-depth structured (MOC) training and (2) use of SEEKonline vs. the Traditional approach to guide fidelity of model delivery on adoption and delivery of SEEK in primary care settings, and subsequent prevention of CM.
Role: Site Principal Investigator

Recently Completed

NIH-NIDA R34 DA 037516-01A1 Palinkas (PI) 4/15/15 – 3/30/19

Measuring Sustainment in Prevention Programs and Initiatives

To design and pilot test a measurement system for monitoring and providing feedback regarding sustainment within four of SAMHSA's prevention-related grant programs.

Role: Principal Investigator

Emory University (Sub), T811267 DiClemente (PI) 7/1/13 – 6/30/18

NIH-NIDA U01 DA036233

KiiDS: Knowing About Intervention Implementation in Detention Sites

To develop implementation strategies that enhance the adoption, fidelity, quality, reach, sustainability and cost-effectiveness of evidence-based substance use and HIV prevention interventions for youth in the juvenile justice system

Role: Site Principal Investigator

NIH-NIDA R21/R33 DA035634 Guerrero (PI) 5/01/13 – 4/30/18

Organizational Capacity to Eliminate Outcome Disparities under Healthcare Reform

To examine the organizational impact of the current expansion of Medicaid on the capacity of community-based OSAT programs to enhance treatment access and retention among minorities residing in Los Angeles.

Role: Co-Investigator

HHS-ACF 90YR0074-01-00 Mennen (PI) 9/30/13 – 9/29/18

Healthy Moms-Healthy Kids: Reducing Maternal Depression for Better Outcomes in Head Start Children

The goal of this project is to test the ability of an evidence-based intervention, interpersonal psychotherapy for groups, to reduce depression among mothers in Head Start to improve parenting and child outcomes.

Role: Co-Investigator

The Oregon Social Learning Center (Sub) 004054-00004, Chamberlain (PI) 1/1/14–4/30/17

NIH-NIDA P50A035763

Translational Research on the Prevention of Drug Abuse and HIV Risk

To understand underlying mechanisms and processes associated with exposure to high levels of early life adversity, and specific to risky decision-making in certain social contexts that are common for CWS youth during early adolescence; develop interventions to reduce high rates of drug use and engagement in HIV-risk behaviors in adolescent girls in the CWS; and identify methods for implementing evidence-based interventions into CWS real-world settings

Role: Site Principal Investigator

PHS Fellowship Supplemental Form

OMB Number: 0925-0001
Expiration Date: 03/31/2020**Introduction**

1. Introduction to Application

(for Resubmission applications)

Fellowship Applicant Section

2. Applicant's Background and Goals for Fellowship Training*

14__Background_and_Goals_For_FT [REDACTED].pdf

Research Training Plan Section

3. Specific Aims*

15_Specific_Aims [REDACTED].pdf

4. Research Strategy*

16_Research_Strategy [REDACTED].pdf

5. Respective Contributions*

17_Respective_Contributions [REDACTED].pdf

6. Selection of Sponsor and Institution*

18__Selection_of_Sponsor__Inst [REDACTED].pdf

7. Progress Report Publication List

(for Renewal applications)

8. Training in the Responsible Conduct of Research*

19__Training_in_Responsible_CR [REDACTED].pdf

Sponsor(s), Collaborator(s) and Consultant(s) Section

9. Sponsor and Co-Sponsor Statements

20_Sponsor_and_Cosponsor_Statements [REDACTED].pdf

10. Letters of Support from Collaborators, Contributors and Consultants

Letters_of_Support [REDACTED].pdf

Institutional Environment and Commitment to Training Section

11. Description of Institutional Environment and Commitment to Training

Description_of_IE__CT1 [REDACTED].pdf

Other Research Training Plan Section**Vertebrate Animals**

The following item is taken from the Research & Related Other Project Information form and repeated here for your reference. Any change to this item must be made on the Research & Related Other Project Information form.

Are Vertebrate Animals Used? ☐ Yes ☒ No

12. Are vertebrate animals euthanized?

If "Yes" to euthanasia

Is method consistent with American Veterinary Medical Association (AVMA) guidelines?

If "No" to AVMA guidelines, describe method and provide scientific justification

13. Vertebrate Animals

PHS Fellowship Supplemental Form

Other Research Training Plan Information

14. Select Agent Research

15. Resource Sharing Plan

16. Authentication of Key Biological and/or Chemical Resources

BiologicalChemicalResources1016533198.pdf

Additional Information Section

17. Human Embryonic Stem Cells

Does the proposed project involve human embryonic stem cells?* ☐ Yes ☒ No

If the proposed project involves human embryonic stem cells, list below the registration number of the specific cell line(s), using the registry information provided within the agency instructions. Or, if a specific stem cell line cannot be referenced at this time, please check the box indicating that one from the registry will be used:

Specific stem cell line cannot be referenced at this time. One from the registry will be used.

Cell Line(s):

18. Alternate Phone Number: 

19. Degree Sought During Proposed Award:

Degree:

If "other", indicate degree type:

Expected Completion Date (MM/YYYY):

PHD: Doctor of Philosophy

05/2021

20. Field of Training for Current Proposal*:

980 Social Work

21. Current or Prior Kirschstein-NRSA Support?*

☐ Yes ☒ No

If yes, identify current and prior Kirschstein-NRSA support below:

Level*	Type*	Start Date (if known)	End Date (if known)	Grant Number (if known)

22. Applications for Concurrent Support?*

☐ Yes ☒ No

If yes, describe in an attached file:

23. Citizenship*

U.S. Citizen U.S. Citizen or Non-Citizen National

Non-U.S. Citizen

If you are a non-U.S. citizen with a temporary visa applying for an award that requires permanent residency status, and expect to be granted a permanent resident visa by the start date of the award, check here: ☐

Name of Former Institution:*

24. ☐ Change of Sponsoring Institution

PHS Fellowship Supplemental Form

Budget Section

All Fellowship Applicants:

25. Tuition and Fees*:

☐ None Requested ☒ Funds Requested

Year 1	\$16,000.00
Year 2	\$17,914.00
Year 3	\$17,914.00
Year 4	\$17,914.00
Year 5	\$17,914.00
Year 6 (when applicable)	
Total Funds Requested:	\$87,656.00

Senior Fellowship Applicants Only:

26. Present Institutional Base Salary:	Amount	Academic Period	Number of Months
27. Stipends/Salary During First Year of Proposed Fellowship:			
a. Federal Stipend Requested:	Amount	Number of Months	
b. Supplementation from Other Sources:	Amount	Number of Months	
	Type (e.g.,sabbatical leave, salary)		
	Source		

Appendix

28. Appendix

Background and Goals for Fellowship Training

A. Doctoral Dissertation and Research Experience

My research interests in understanding the aging process and improving the quality of life for older adults were first developed in the three-year research assistant experience in the lab of aging and cognition at Zhejiang University. I applied for the Zhejiang Province Undergraduate Innovation and Entrepreneurship Fellowship and was funded to test the reliability and feasibility of using an internet administered 9-item Patient Health Questionnaire (PHQ-9) as a depression screening tool for Chinese older adults. Furthermore, we developed a cellphone application, integrating the function of self-diagnosis, mood monitoring, and online intervention materials. The mobile app encompasses Problem Solving Therapy courses filmed by our team to improve patients' access to therapy resources. I was drawn to explore the potentials of employing information and communication technology (ICT) to improve older adults' quality of lives through the first experience of using mobile technology to deliver mental health services. This research experience produced a paper published in *Asia-Pacific Psychiatry*.

As a first-generation college student at Zhejiang University, I seized as many opportunities to get research training as I possibly could. I found gratification in building new knowledge and was fascinated by the possibilities in making positive change using research. I visited the University of Surabaya to collaborate with researchers in Indonesia to conduct cross-culture research. I worked with Dr. Christ Mayhorn on the topic of evaluating the social engineering of phishing attacks at North Carolina State University. I was selected as one of the three undergraduates at Zhejiang University to work as a paid research intern at the University of Alberta at the School of Nursing. During the three-months internship, I worked with Drs. Carole Estabrooks and Matthias Hoben on the Translating Research in Elder Care program. I studied how does the working environments of nursing homes affect the physical and mental health of its employees. I also participated in a systematic review on the topic of the barriers and facilitators of oral health care for older adults residing in nursing homes, which was published in the *International Journal of Nursing Studies*. My experiences at multiple cities in China, Indonesia, Canada, and the United States exposed me to the rich cultures at diverse regions of the world, which inspires and empowers me to research with populations with diverse backgrounds. Through working on aging-related research topics, I realized the vast unmet needs in research and practice with the older adult population in my China and globally. I decided to commit my research to this vulnerable group with complex challenges.

By the end of my undergraduate training, I had a strong aspire and a clear vision of becoming an investigator in applied science. Therefore, I applied for the MSW/PhD dual degree program at the University of Southern California Suzanne Dworak-Peck School of Social Work (USC SSW). Located in central Los Angeles, USC provides a unique research environment for studying older adults with diverse cultural backgrounds and linguistic preferences. Driven by my interest in employing technology to improve older adults' health outcomes, I participated in the Intergenerational Mobile Technology Opportunities Program (IMTOP) led by Drs. Shinyi Wu and Iris Chi. The Intergenerational Mobile Technology Opportunities Program (IMTOP) is a mHealth intervention that recruited college-student tutors to help older patients to learn how to use mobile technology and a health tracking app to facilitate diabetes self-management. Using the data collected in the IMTOP intervention study, my first-authored paper reporting the longitudinal IMTOP intervention effects in rural Taiwan won the 2018 APHA Nobuo Maeda International Research Award. The manuscript was submitted to *The Diabetes Educator* and is currently under review. Additionally, I presented more results from the IMTOP study at annual scientific meetings of the Society of Social Work and Research and Gerontological Society of America. The hands-on research activities conducted with IMTOP not only helped me learn substantial knowledge in designing and implementing mobile health (mHealth) interventions but also gained excellent training in theoretical thinking and scientific writing. Furthermore, IMTOP study leads to two more research opportunities in which I contributed significantly – IMTOP pilot study for immigrant populations in Los Angeles and a feasibility study of an intergenerational mHealth program for affordable housing communities (ImPAHC) in Sacramento.

Encouraged by the promising intervention effects found in Taiwan, we decided to adapt the IMTOP intervention to meet type 2 diabetes management needs for immigrants in Los Angeles. I led IMTOP pilot study to test the feasibility of using the technology-facilitated diabetes self-management intervention to address the prevalent problem of type 2 diabetes (T2D) among first generation Chinese and Hispanic immigrants. The research project adopts a mixed-method approach, using both survey and individual semi-structured interviews to collect data. The research project is a collaboration between USC Edward R. Roybal Institute on Aging and four community-based organizations serving recent Chinese and Hispanic immigrants. I paid weekly visits to all four community collaborators and communicated with managers and staff members of the organizations to formulate our research plan. I supervised 3 MSW students and one undergraduate student and provided training in qualitative research methods and questionnaire administration. Working together, we have collected 140

questionnaires with recent immigrants with type 2 diabetes and conducted 24 in-depth individual interviews with participants who speak Cantonese, Mandarin, and Spanish. Because of this pilot study, I had the opportunity to work closely with community-based organizations and gained experience in data collection and research program management. The IMTOP pilot study yields two manuscripts under preparation.

The results of the IMTOP pilot study highlighted the importance of social, cultural environments on health outcomes. Collaborating with community-based organizations significantly increased our access to the target population of the intervention and provided future avenues for disseminating the research results. Employing what we learned in both IMTOP and IMTOP pilot study, Dr. Wu, Dr. Chi, and I submitted a proposal to the USC SSW Adult Mental Health and Well-being (AMHW) department and were awarded a pilot funding for exploring the feasibility of delivering a mobile health facilitated T2D intervention at affordable retirement communities (ImPAHC). I served as a co-investigator on the ImPAHC study. We paid visits to three retirement communities and conducted semi-structured interviews with professionals and staff and had focus group discussions with older adult residents. I am working with two other graduate students to analyze the qualitative data with thematic content analysis methods. Working with retirement housing communities inspires me to investigate the aging process and social determinants of health through the lenses of ecological theory, which leads to my decision of investigating the physical and social environments in my dissertation research.

Besides research opportunities provided in the doctoral program, the MSW training of the dual degree program provided opportunities for me to gain clinical practice experience and work directly with older adults in my field placement. I interned at ABC Adult Day Health Care Center for 20+ hours per week during my first year in the MSW/PhD dual degree program. In my clinical practice, I witnessed the prevalent loneliness and social isolation among seniors and observed the strong wishes of older adults in getting connected with family and friends. Social isolation and loneliness have become the primary outcomes in my research since then. Many older adults came to consult me on how to use their mobile phones to contact friends and sharing meaningful memories online with their families. My experience working closely with older adults affirm the value of addressing social isolation by bringing information technology to older adult's life.

My clinical practice and research experiences made me aware that the majority of the currently available technology tools, such as social media sites, mobile health apps, are not designed with older adults' unique needs in mind. Older people often are not included as part of the intervention/technological tool development team, which adds barriers for older people to learn to use technology. To address this issue, I serve as a co-investigator on a research project, led by Drs. Wu and Chi, employing a participatory co-design approach to develop a caregiver self-management program (CSMP) app for the Chinese immigrant caregivers of older adults. The co-design CSMP app proposal was funded by the USC Clinical and Translational Science Institute (CTSI) and AMHW pilot awards. We have completed the conceptual design, interactive prototype testing, and internal real-life condition testing phases of the co-design process. I am currently working with a graduate student colleague to conduct qualitative data analysis and drafting the manuscript reporting the unique needs of the Chinese immigrant caregiver population and their acceptance of technology-facilitated interventions. The CSMP app co-design study preliminarily supported the feasibility of delivering a traditional in-person intervention with a mobile technology platform. I also gained first-hand experience using the co-design process for intervention tool development and pilot research.

I pursued my interest in studying social isolation and loneliness in directed research tutorials and my qualifying exam project. Using data from the Health and Retirement Study (HRS), I examined the longitudinal association between internet use and loneliness for older people. The manuscript prepared with this work is currently in revision and resubmission. Because of my strong interests in statistical methodology, I sought training in structural equation modeling (SEM) and longitudinal data modeling at the Inter-university Consortium for Political and Social Research (ICPSR) summer training at the University of Michigan. Learning advanced statistical methodology enabled me to model the relationships of key variables of interests in a sophisticated manner. I further honed my data management and modeling skills in the process of conducting independent research with the qualifying exam. The qualifying exam project examined the longitudinal bidirectional relationship between the accumulation of geriatric conditions and loneliness. The study results were rich enough to be reported in two separate manuscripts that will be submitted to top gerontology journals.

Socially isolated older adults are at high risk of cognitive decline and the onset of Alzheimer's disease and related dementias (ADRD). My past research experiences shaped the path for me to study the longitudinal association between environmental factors, social isolation, and cognitive health. With the awareness that cognitive health in later life is a new research field for me, I sought the highly individualized training opportunities that could be made possible with the F99/K00 award. If awarded, this funding opportunity will enable me to

receive exceptional training and mentored research, transition my research directions and help me grow into an independent investigator conducting social isolation and cognitive health research from an ecological perspective.

B. Training Goals and Objectives

My research is guided by the overarching goal of promoting long, healthy, and meaningful lives. My long-term career development objective is to build substantial knowledge that leads to effective interventions from a social-ecological/community engagement perspective to reduce social isolation and prevent cognitive decline. I envision myself to be a leading scientist in a renowned research-oriented university and making impacts with my research on social isolation and cognitive health. For the F99/K00 award tenure, I aspire to gain knowledge in cognitive health and ADRD research, receive training in analyzing high-frequency digital biomarkers longitudinal data and neuroimaging methodologies, and expand experience in intervention development, implementation, and assessment. Working with my sponsors, I developed the following specific training goals and objectives and described how we plan to collaborate to achieve the training goals.

Training Goal 1: increase knowledge in prevention, intervention, and policy pertaining to cognitive decline in normal aging and the development of ADRD. I will work with my co-sponsors, Dr. Iris Chi, Dr. Shinyi Wu, and Dr. Hiroko Dodge, to get immersed in cognitive health research. I will systematically learn the protective and risk factors of cognitive decline, currently available treatment, and interventions for ADRD through mentored research. I have already been actively engaged in the academic activities at the USC Edward R. Roybal Institute on Aging and have been attending research meetings where frontier aging and ADRD related research findings are presented. With the receipt of the F99/K00, I will get involved with the Alzheimer's Disease Research Center (ADRC) at USC and the Layton Aging and Alzheimer's Disease Center at Oregon Health and Science University (OADC). Dr. Wu will help me to get connected with researchers at USC ADRC. I will learn new progress in ADRD research by getting engaged with USC ADRC and building collaborative relationships for future research. Dr. Dodge will be the liaison for the resources at OADC, directing me to the personals and research cores according to my learning needs. Dr. Dodge and Dr. Wu will provide directed reading to facilitate my learning for substantial topics related to cognitive decline and ADRD. I will present and discuss the papers with peers and mentors in our regular journal club hosted by Dr. Wu. Additionally, I will join through Zoom the monthly OADC investigator meeting held at OHSU where researchers present to critique their ongoing research and proposals. I plan to take a course at the USC School of Gerontology dedicated to discussing special issues on cognitive decline and ADRD. This course covers cognitive decline for people with Alzheimer's Disease or

The new knowledge in cognitive health gained through the F99 phase will enable me to conduct informed cutting-edge research at the K00 phase and beyond.

Training Goal 2: gain skills in analyzing high-frequency digital biomarkers longitudinal data and employing neuroimaging methods. Through conducting research with the IMTOP study data, I had some experience in utilizing biomarker measures, such as blood pressure and blood glucose, yet I have not systematically learned about assessing and analyzing biomarker data in ADRD related research. Cognitive health studies often require the use of bio-marker indicators to objectively evaluate the intervention effect or longitudinal changes in brain functioning. Neuroimaging methodology is widely used in ADRD related research, which allows the researchers to link the variation observed in behavioral testing of cognition with activities in the brain. Learning to assess biomarkers as key variables in my research and using neuroimaging methods will enable me to have in-depth discussions with a broader scientific audience, including but not limited to researchers in fields of Social Work, Neurology, Gerontology, and Engineering. Using biomarker and neuroimaging methodology will also equip me to build knowledge on detecting early-stage brain changes related to cognitive decline, which can produce scientific evidence that informs social work intervention and practice with the socially isolated older adults. I plan to receive training conduct hands-on research in Neuroimaging methods in the K00 stage, at the OADC Neuroimaging core. The OADC is also specialized in collecting digital bio-marker data. Digital biomarkers are collected using wearable devices in a nonintrusive manner, such as walking speed assessed using in-home sensors. Enhancing my capacity of understanding and using digital biomarker data will expand the spectrum of outcomes I can investigate, including those indicators that are essential for ADRD research.

I will hone my longitudinal data analysis skills in the process of analyzing four waves of longitudinal data from the NHATS study, which prepares me to analyze high-frequency digital biomarker longitudinal data in the K00 phase. I will complete a self-directed course on using R programming to handle large longitudinal datasets and run sophisticated models that is suitable for answering my research questions, such as multi-level modeling, structural equation modeling, and Bayesian inference models.

Training Goal 3: learn to design, implement, and evaluate community-based interventions. In the F99 phase, I will apply for a pilot funding at OADC with Dr. Dodge to conduct focus groups with participants from her on-going R01 project “Internet-based Conversational Engagement Clinical Trial (I-CONNECT)” (NIA R01AG056102, R01AG056628 & R56AG051628). In this qualitative pilot study, we will explore with the participants a potentially sustainable approach to increase the social engagements among isolated older adults. Data collected in the pilot study will complement my quantitative longitudinal analysis using the National Health and Aging Trend Study (NHATS) dataset. I will gain practical experience in observing field works of I-CONNECT, working with Dr. Dodge and the I-CONNECT team as they enroll participants, set up homes with chrome books and internet, and conduct conversational sessions as well as end the trial, and provide assistance after participants completed the trials. I will observe how the team monitors recruitment, enrollment, and randomization. This series of observation/shadowing will give me the hands-on experience of conducting non-pharmacological trials in preventing social isolation and cognitive decline using modern technology.

Mixed-methods research is uniquely advantaged in obtaining a holistic perspective of the problem targeted by the intervention within the context, which is essential in developing quality interventions. I will work with Dr. Lawrence Palinkas to formulate the purposeful sampling plan and design the focus group interview guide. I will also receive direct qualitative data analysis training from Dr. Palinkas through regular meetings. Dr. Palinkas is also specialized in implementation and dissemination research. I will consult Dr. Palinkas on the design of evaluating the implementation process of the intervention after a draft of the community-based intervention for social isolation, and cognitive health has been developed.

Dr. Chi has excessive experience working with community-based organizations that mainly serve the older adult population. I am serving as a co-investigator on Dr. Chi's research project, studying caregivers' self-care behavior at an affordable housing facility for older adults (Golden Age Village, GAV). With the support of this award, I will be further involved in intervention research at GAV and receive training on conducting community-based research. Besides, I am engaged in a mHealth research project lead with Dr. Wu in collaboration with Life Steps foundation dedicated to providing the health, psychological, and socioeconomic services to older people. I have been working closely with my co-sponsor Dr. Wu to harness the capacity of mobile technology to promote self-management of chronic disease among older type 2 diabetes patients. With the support of the F99/K00 award, I will continue to conduct research in collaboration with Dr. Wu to gain understanding of the role ICT to enhance health and social connectedness of older adults. I received a travel fellowship to attend the *Forming Science-Industry Research Collaboration Workshop* organized by the Stanford Longevity Center at the 2019 GSA annual science meeting. These training opportunities will allow me closely work with community-based organizations and will train me to develop, adapt, and evaluate interventions in community settings. The experience working directly with organizations will facilitate me to achieve the long term goal of becoming an impactful intervention researcher. Firsthand understandings gained from the community-based research experience will also be incorporated in my teaching experience during the tenure of the award, which will prepare me to become a future faculty member in a reputable academic institution.

Receiving the F99/K00 award will provide me the opportunities to have invaluable guidance from leaders and experts in the fields of Gerontology, Engineering, Neurology, Mixed-methods research, and Dissemination and Implementation Science. Applying to the F99/K00 award encouraged me to think beyond the dissertation stage of research and identified a postdoctoral mentor match with my career development and training needs in advance. If awarded, I will be able to work with both my doctoral mentors (Chi & Wu) as well as my postdoctoral mentor (Dodge) simultaneously and receive individualized training working with Dr. Palinkas. Having a smooth transition from the predoctoral training to the postdoctoral self-directed research will benefit the realization of both my short and long term career development goals by increasing the efficiency and continuity of the transitional stage and enhance the academic collaboration with faculty members of similar research interests outside USC SSW.

C. Activities Planned Under this Award

During the proposed five years of F99 and K00 phases of training, the majority (75%-85%) of efforts will be dedicated to research activities, including conducting literature reviews, data collection and analysis, and preparing manuscripts and submit to top peer-reviewed journals in the fields Gerontology and Neurology. Besides research activities, in the F99 phase, I will co-teach a doctoral-level statistics course. Teaching experience is essential to prepare me for an academic position in the future. As I develop from a PhD candidate to a postdoctoral fellow, I will spend more time (5%) on mentoring students in preparation for becoming a faculty member and independent researcher.

In the F99 phase, I will take a course on cognitive decline from USC Davis School of Gerontology (GERO-566) to enhance my knowledge for AD/DRD related topics. I will complete a formal online course on Study Designs

for Intervention Research in Real-World Settings at UCSF. This course will help me to gain knowledge in determining which design is most suited to a range of 'real-world' implementation settings and circumstances, and on choosing between designs to maximize overall study quality. In the K00 phase, I will apply for the NINDS-sponsored Clinical Trial Methodology Course (R25NS088248), a two-year intensive, engaging program with online and in-person training sessions designed to help junior investigators develop scientifically rigorous, yet practical clinical trial protocols, and to focus funding mechanisms as a key trial planning activity. I will complete a self-directed course on useful R programming skills, including loading and manipulating data within R, simple visualization, data quality control, and statistics. In Year 1 of the K00 stage, I will apply to attend a week-long Health and Retirement study summer institute hosted University of Michigan Institute of Survey Research to deepen my knowledge of social determinants of health. Coursework will make up about 10% of my total professional efforts in the F99 phase and the first year of the K00 phase, about 5% of the effort in the second year of the K00 phase.

Percentage of efforts

	F99-Year 1	K00-Year 1	K00- Year 2	K00- Year 3	K00- Year 4
Research	75%	80%	80%	80%	80%
Mentoring	0%	5%	5%	5%	5%
Teaching	5%	0%	0%	0%	0%
Coursework	10%	10%	5%	0%	0%
Grant writing	5%	0%	5%	10%	5%
Conferences	5%	5%	5%	5%	10%

I will work with my co-sponsors to continuously apply to funding opportunities from NIH and/or other prestige funders to support the development and pilot testing of social isolation and cognitive decline interventions. I will engage in peer-support writing groups, complete seminars in responsible conduct in research, and attend the "Design Studio" and Advanced Grant Writing Course at OHSU in order to refine my future career development and/or research award proposal to be submitted during the K00 award period. Grant writing will make up for about 5% to 10% of my total efforts. More time will be spent on grant writing in year 3 of the K00 phase because I will be preparing for a submission for another career development award.

Knowledge developed from the proposed dissertation will be presented at the annual meetings of the GSA, APHA, and SSWR. I am a member of all three scientific organizations and have been actively participated in the aging practice and research, aging & public health, and social isolation special interest groups for three years. I will continue to make scholarly contributions to the organization and groups with new knowledge and skills gained from my proposed research projects. In the K00 phase, I will expand my network to AD RD related research groups, I will attend and submit abstracts to present at the Alzheimer's Association International Conference (AAIC), GSA and/or Clinical Trials in Alzheimer's Disease conference (CTAD). I will also work with my mentors to identify 2-3 additional scientific meetings that will appropriately expand my collaborations and network. I will have conversations with distinguished researchers in my research area and foster long-term collaborative relationships in the academic community. Conferences takes about 5% of my efforts except for year 4 of K00. More efforts (10%) were budgeted considering I will be seeking for a faulty position in the last year of the award tenure.

My short term career milestone is to graduate with distinction from USC SSW. Within four years after finishing the dissertation, I envision myself successfully attained another career development award and hold a tenure-track assistant professor position at a research-intensive university. Nine years after I complete my dissertation, I envision myself achieving tenure and making impacts through my research on social isolation and cognitive health with older adults.

Training activities and timeline

Training activities and timeline						
Training activities	Sponsor/Professor	F99	K00			
		Year 1	Year 1	Year 2	Year 3	Year 4
Goal 1: increase knowledge in prevention, intervention and policy pertain cognitive decline in normal aging and the development of ADRD						
Developing literature review database on cognitive health and ADRD research	Chi & Wu	X				
Regular consultation on topics in cognitive health and ADRD research	Dodge	X	X	X	X	X

Course: GERO-566 Cognitive Decline: Alzheimer's Disease and Dementia	TBD	X				
Get involved with research activities at USC ADRC and OADC - Network with researchers with similar interests at USC ADRC - Join monthly OADC investigator meetings (Online at F99 phase, in-person at K00 phase)	Wu & Dodge	X	X	X	X	X
Goal 2: gain skills in analyzing high-frequency digital biomarkers longitudinal data and employing neuroimaging methods.						
Directed reading in cognitive health research using neuroimaging and biomarker outcomes	Dodge & Wu	X	X			
Attend Neuroimaging trainings and seminars at OADC	Dodge & Silbert		X	X		
Learn to incorporate biomarker and neuroimaging outcomes with hands-on research activities	Dodge		X	X	X	X
Self-directed course on using R programming skills	NA		X			
Goal 3: learn to design, implement and evaluate community based interventions						
Collect pilot data to inform intervention design	Dodge	X				
Immersion training with I-CONNECT field work, observe how the team monitors recruitment, enrollment, and randomization.	Dodge		X	X		
Regular consultation on mixed-methods research design, implementation and data analysis	Palinkas	X	X			
Regular consultation on intervention implementation and dissemination science	Palinkas			X	X	X
Gain experience in conducting collaborative research with community-based organizations by participating in research projects	Chi & Wu	X	X	X	X	X
Online course: Study Designs for Intervention Research in Real-World Settings (UCSF)	TBD	X				
NINDS-sponsored Clinical Trial Methodology Course (R25NS088248)	TBD		X	X		
Dissertation and Professional Development Activities						
Prepare and submit conference abstracts	All sponsors	X	X	X	X	X
Attend conferences and present research works	NA	X	X	X	X	X
Preparing manuscripts and submit to high impact peer-review journals	All sponsors	X	X	X	X	X
Gain further grant writing experiences and prepare to submit another career award application	All sponsors			X	X	X
Teaching	NA	X				
Mentoring students	NA		X	X	X	X

Specific Aims

Social well-being and cognitive health are two essential components of successful aging.¹ Three decades of research have consistently shown that social isolation is associated with the onset of cardiovascular disease,²⁻⁴ declines in cognitive function,⁵⁻⁷ higher risks of depression and anxiety,⁸ and lower overall life satisfaction.⁹⁻¹¹ Perceived social isolation was associated with a higher cortical Amyloid burden in older adults, supporting the crucial role of social isolation in preclinical Alzheimer's disease and related dementia.¹² The potential causes of social isolation are multi-faceted. While the existing literature studied individual-level factors that were associated with social isolation, such as age,^{13,14} personality¹⁵⁻¹⁷ and health status,¹⁸⁻²¹ only a few studies have examined the extent to which the living environment influences the experience of social isolation among older adults.

The Ecological Theory of Aging (ETA) posits that the living environment profoundly influences the aging process.^{22,23} The process of personal and environmental resource exchange determines the aging outcomes through building their identity, evoking or reducing stress, and maintaining or hindering the autonomy of an older person.²² Environments not only affect social well-being of older adults, but it also serves as one of the most pervasive and complex stimuli to the brain and contribute to shaping cognitive functionality through the mechanism of neuroplasticity.²⁴ Socially isolated individuals are at higher risk of cognitive decline.^{25,26} Social isolation might mediate the association between environments and cognitive health. The proposed dissertation work aspires to investigate the longitudinal effects of multiple dimensions of the living environments on social isolation and cognitive health for older adults, including physical environment measured by community and in-home disorder and social environment measured by neighborhood cohesion.

Information and Communication Technology (ICT) has brought a revolution to the way people communicate and access resources.^{27,28} With increased popularity among older people,^{29,30} ICT provides excellent opportunities to address social isolation and promoting cognitive health.³¹ An increasing amount of research investigating the association between ICT use and social isolation have been published, however, whether ICT use would interact with the effect of environmental factors on the risk of social isolation has not yet been empirically tested.³¹ Because ICT increases access to information and resources, adopting ICT possesses the potential to mitigate the effect of a disadvantaged living environment.

For the F99 phase, I will examine the longitudinal impact of the environments of living on social isolation and cognitive health with four waves of data from the National Health & Aging Trend Study (NHATS). The subjective perception of the environmental influence on social well-being will be elicited using focus groups. With the older adult participants' input, I will explore the possible designs of a community-based, technology-facilitated approach that could sustainably enhance the social well-being and cognitive health of isolated older adults.

Cognition health is the foundation for independent living and quality of life. The effect of ICT use on cognitive health has not been extensively researched. Promoting social interaction using technology-based approaches can alleviate the social isolation among older people and prevent cognitive decline at the same time, yet only a few randomized controlled intervention studies have been conducted to examine the effectiveness of ICT-enhanced tools on cognitive functioning and social isolation.³² The existing trial showed positive effects for ICT-enhanced intervention.

At the K00 phase, I will use the longitudinal data from the ongoing randomized controlled trial "Internet-based Conversational Engagement Clinical Trial (I-CONNECT)" (NIA R01AG056102, R01AG056628 & R56AG051628) to investigate the mechanism of conversational engagement on the cognitive functioning. The I-CONNECT trial is currently collecting digital bio-marker and neuroimaging data, including medication adherence, speech and language characteristics, and region-specific structural and functional pre-post changes in the brain. Using this rich data set, I plan to examine why and how social interaction affects the cognitive functioning of older adults, which is not well understood in literature, as well as how technology can be used effectively to alleviate isolation and improve cognitive functions.

Aim 1: In the F99 phase, I will examine the effect of physical and social environments on social isolation and cognitive health using a mixed-methods approach. ICT use is hypothesized to moderate the influence of physical and social environments.

Aim 2: In the K00 phase, I will examine the effectiveness of an ICT-enhanced conversational intervention on social isolation and its effect on cognitive health among older adults with normal cognition or mild cognitive impairment (MCI). I will explore the potential to co-design a technology-facilitated community-based intervention from an ETA perspective for social isolation and cognitive decline with older adult participants.

Successfully achieving the aims of this proposal will produce knowledge on the environmental factors on the risk of social isolation among older adult, contribute to the existing literature on the mechanisms of social engagement on cognitive health, and lead to the development of a technology-facilitated intervention program for social isolation and cognitive decline from the perspective of ETA.

Research Strategy

1. Significance

1.1. Environment and Social Isolation

Social isolation is the absence of contact with other individuals and a lack of relationships and social integration.³³ Empirical evidence repeatedly showed that social isolation is associated with death by cardiovascular disease and early morbidity among older adults.²⁻⁴ A meta-analysis with 70 studies published between 1980 and 2014 showed that socially isolated individuals had a 29% increased likelihood of mortality.⁴ The research and practice fields have increasingly recognize the significance of addressing the problem of social isolation. AARP estimated that about 4 million older adults enrolled in Medicare are socially isolated, and Medicare spends an additional \$6.7 billion on isolated individuals than otherwise if they are socially connected.³⁴ Eradicate the problem of social isolation requires a social approach. However, while the current literature has explored the individual-level risk and protective factors of social isolation, such as personality and health concerns, how does the physical and social characteristics of the living environment affects the experience of social isolation has not been well researched.

The Ecological Theory of Aging (ETA) posits environmental factors have significant influences on health and social wellbeing.^{22,35,36} Living environment provides access to resources, and the settings for social interaction as well as an essential source of cognitive stimulation.²⁴ Living environments are often assessed through diverse dimensions including physical and social aspects of environments.^{22,36} Physical environments refer to the objective aspects of the neighborhood and the facilities, for example, the safety of the neighborhood, access to amenities, green open spaces and aesthetics.³⁷ While social environment often been operationalized as the social participation and social integration perceived by people, such as participate in social activities, sense of community, and trust in people.³⁸

For the association between **physical environmental factors and social isolation**, previous research reported that older residents of a high-crime neighborhood are more likely to be socially isolated despite their strong desire for social integration.^{8,39} The community disorders, which can present itself as litters on the streets, deserted buildings, and vandalism, could lead to the perception of lack of safety, thus causes increased likelihood of social isolation. Outdoor space & public buildings, and common areas have a critical role in keeping the residents living an active social life and contribute to their general wellbeing.³⁷ Although existing literature shed light on the role of the physical environment on social isolation, the evidence is built mainly on cross-sectional data, the longitudinal association between the two warrants further research. The proposed dissertation work aspires to build evidence with longitudinal data, and the following hypothesis was derived based on cross-sectional results and theoretical reasoning of ETA.

H1: Negative physical characteristics of the environment, i.e., disorders in the community and at home, predicts the increased social isolation in the community.

Social environmental factors like community support & health services, communication & information, civic participation, & employment have been empirically supported as protective factors of social isolation.^{40,41} Sense of belongingness, togetherness, and trust were found to be associated with a less sedentary lifestyle and more social participation and interpersonal interaction, and thus, predicts a lower extent of social isolation.⁴² Residential satisfaction and sense of belonging was found negatively associated with perceived social isolation.⁴³

H2: Positive social environment factors (e.g., trust neighbors, people in the community, helping each other) are associated with less social isolation longitudinally.

One's perception of the social environment is likely to be influenced by the physical characteristics of the environment. Sampson and Raudenbush (2004) reported that physical deterioration and disorder has a social meaning beyond just the observation of neighborhood disorder.⁴⁴ Physical characteristics of a living environment could change an individual's opinions of the social environments, resulting in less perceived opportunities in the neighborhood. For example, visible vandalism on the street could prevent residents from walking on the streets, which could increase the incidence of social isolation.⁴⁵

H3: Social environmental factors mediate the longitudinal association between physical environmental factors and social isolation.

Why is it that the social and physical aspects of the living environment affect the social wellbeing of its older residents? This question remained unanswered with previous research. ETA provided three propositional processes for the environment to "get under the skin", i.e., autonomy defined as the older adult take action to change the environment to fit their needs, belonging defined as a sense of security, inclusion and connectedness, and stress defined as the "environmental press" to which made the individual suffer.^{22,36} Yet the corresponding measures are not available in nationally representative longitudinal studies. A qualitative exploratory approach will be employed in this proposed dissertation work to elicit older adults' own experiences

on the impact of the living environment for the experience of social isolation. ETA will be used to guide the design of the interview guide. No hypothesis was derived for the qualitative inquiry with exploratory nature.

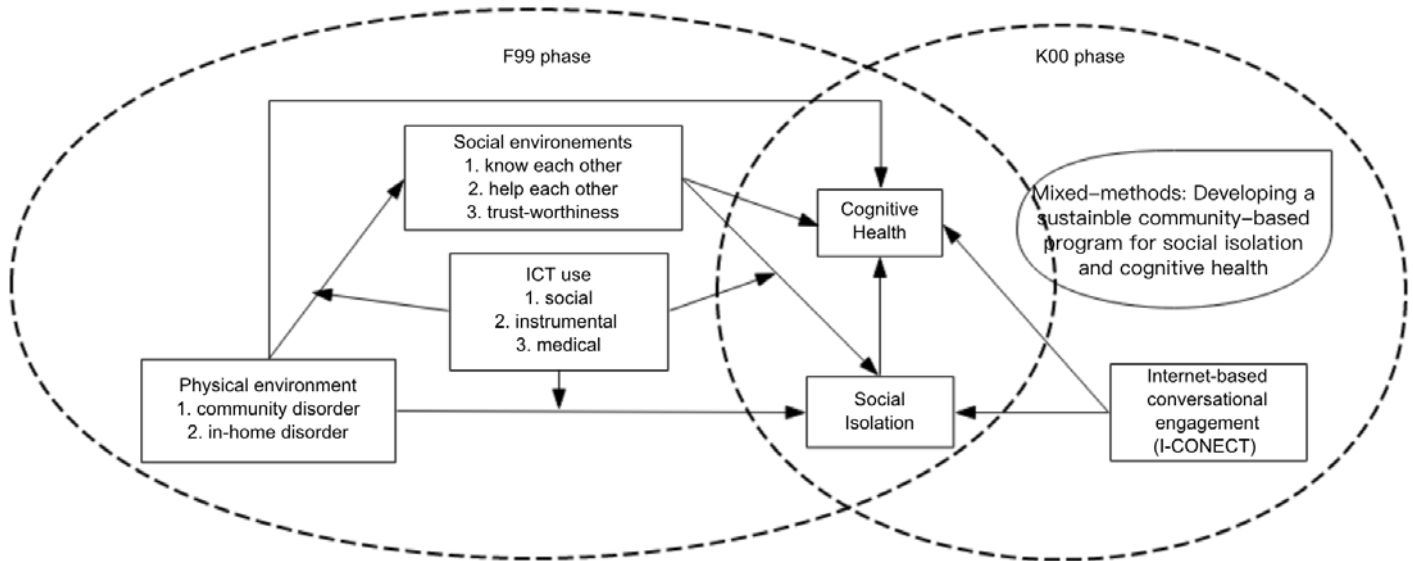


Figure 1. Conceptual framework of the F99 and K00 phases of research

1.2. Environment, Social Isolation, and Cognitive Health

Physical and social environments not only affect the social wellbeing of older people but also fundamentally influences their cognitive health. Environments serve as one of the most pervasive and complex stimuli to the brain and contribute to shaping cognitive functionality through the mechanism of neuroplasticity.²⁴ Environmental challenges, as simple as daily tasks like shopping, was hypothesized to simulate the brain because it requires an older person to remember the route and shopping list despite the distractors in the environments. Environmental stimuli, just like education and cognitively-demanding jobs, were theorized to be associated with increased cognitive reserve and prevent cognitive decline.²⁴ Considerable amount of research/practice interests and effects have been paid to building aging-friendly communities,^{46,47} however, little is known about the cognitive effects of the physical and social environments.²⁴ **Carefully investigating specific aspects of visual simulation in the living environment, such as disorder at home and in the community, is granted.**

Community disorder, such as cluttered and unsafe environment, increase the cognitive burden and the environmental stressors on the residents. Older adults living in communities with a higher number of adverse physical environmental characteristics are more likely to be deprived of health care resources and cognitively stimulating activities.⁴⁸ Positive social environments provide the resources and opportunities that are necessary for cognitive simulating engagements.^{38,49} Social environments were found to be significantly associated with cognitive functioning with cross-sectional data.⁴⁹ **Although the theoretical reasons supporting the association between specific physical and social environments and cognitive decline is substantial, empirical examinations of the relationships are scarce.**^{24,50} The proposed research project aspires to contribute to the knowledge by building empirical evidence on physical environmental factors and cognitive health using longitudinal data. The following hypotheses were derived.

H4: Negative physical environments, i.e., the disorder in community and homes, were longitudinally positively associated with deterioration in cognitive function.

H5: Positive social environment, i.e., neighborhood cohesion, longitudinally associated with deterioration in cognitive function.

Social isolation could mediate the association between physical and social environments and cognitive functioning. The literature review in the previous section established the reasoning for the associations between physical and social environments and social isolation. Social isolation was found to predict cognitive decline longitudinally.²⁵ Perceived social isolation was associated with a higher cortical Amyloid burden in older adults, supporting the crucial role of social isolation in preclinical Alzheimer's disease and related dementia.¹²

H6: Social isolation mediates the longitudinal association between physical and social environments and cognitive function.

Although ETA highlights the importance of the environment in the aging process in general, the theory did not specifically discuss the effect of environments on cognitive health.^{22,23} The knowledge that will be built in the proposed study could enrich the content and expanding the scope of ETA.

1.3. ICT, Social Isolation and Cognitive Health

ICT use has been found associated with reduced social isolation.^{51,52} A systematic review reported that ICT alleviates the elderly's social isolation through four mechanisms: connecting to the outside world, gaining social support, engaging in activities of interests, and boosting self-confidence.⁵² The ETA hypothesized ICT use promotes the wellbeing of older adults through enhancing their agency.³⁶ However, the effect of ICT interventions on social isolation were not able to last for more than 6 month after termination.⁵² Previous research noted that the ICT intervention might not work for everyone in the diverse population of older adults with heterogenous personal characteristics and life experiences.⁵² Future research are needed to identify how to implement and contextualize ICT interventions in the older adult's daily life so that the population would be more likely to benefit from the ICT use.⁵² Additionally, the nature and content of ICT use might result in different effects. Internet use for social purposes was reported to be associated with decreased perceived social isolation in the following year, while the association between social isolation with instrumental use of internet was insignificant.⁵³ **Contextualizing the effect of ICT use within the ecological living environment of older adults could contribute to the literature on what kind of ICT is effective in alleviating social isolation under what circumstances.**

The impact of ICT use within the context of the living environment will be explored for the first time in the proposed dissertation work and postdoctoral research. ICT adoption has the potential to reduce the negative influence of disadvantaged physical and social environments by providing a new avenue for older adults who suffer health issues to interact with family, peers, and society.^{27,28} ICT users reported that internet made it easier to reach people and increased the quantity of communication with others, thus supporting them in feeling less isolated.⁵¹ ICT corresponds to the information & communication aspect of the community environment, we hypothesize the effect of physical and social environment on social isolation is different for ICT users and non-ICT users. In the F99 phase, I will investigate the interacted influence of physical and social environments and using ICT for different purposes (social, instrumental, and medical) on social isolation.

H7: ICT use will moderate the longitudinal association between physical and social environments and social isolation experienced by older adults. More specifically, individuals who use ICT for social purposes are less likely to be socially isolated.

The effect of ICT use on cognitive health has not been extensively researched and well-understood. Besides, ICT is an umbrella term summarizing a variety of technological tools that could facilitate interpersonal interaction. I-CONNECT intervention employs a Randomized Controlled Trial (RCT) study design to investigate the effect of a specific type of ICT use, i.e., Internet-facilitated conversational engagement via video-chat, on cognitive functioning among older adults who are cognitively intact or with Mild Cognitive Impairment (MCI). The task of conversation is highly cognitively simulating, which requires control of attention, working memory, executive functions, and mental interpretation of other's emotions and intentions.^{54,55} Linguistic ability is known as highly related to cognitive health among older adults.⁵⁶ In the K00 Phase, I will utilize data collected in the I-CONNECT project to examine the longitudinal intervention effect of ICT facilitated conversation on the social wellbeing and cognitive health of older adults.

The current I-CONNECT intervention did not explicitly take into consideration the impact of physical and social environments. With the evidence will be built in the longitudinal secondary data analysis and qualitative focus groups of in the dissertation, I will explore the possibility of designing a sustainable community-based program to offer a technology-facilitated solution to alleviate social isolation and prevent cognitive decline.⁵⁷ Studying the effectiveness of technology in the living environment and explore new approaches of technology-facilitated care could transform the older adults' home to become more than a living space, but also a service platform.⁵⁸

Figure 1 illustrates the conceptual framework for both F99 and K00 phases. Examining the physical and social environmental impacts on social isolation and cognitive health among older adults, this the proposed research has the potential to produce a sustainable solution to the challenge of social isolation among older adults and the increased likelihood of cognitive decline and ADRD onset accompanies with it. By contextualizing the role of technology in older adults' living environments, knowledge will be built in the proposed research will address the challenges of independent living and aging in place with technology tools that fit the circumstances. Although the distinction between physical and social environments was proposed by the ETA, empirical evidence has examined the subjective aspect of the living environment (i.e., social environment) less adequately.²² The proposed work will contribute to the literature and expand the scope of ETA by studying both the physical and social environments and their relationships with social isolation and cognitive health.

2. Approach

2.1. F99- predoctoral stage

2.1.1. Quantitative inquiry using NHATS data

2.1.1a. Dataset

The proposed dissertation will use data from NHATS wave 5 (collected in 2015) to 8 (collected in 2018). NHATS is a longitudinal study that annually surveys Medicare beneficiaries aged 65 and above in the contiguous United States (excluding Alaska, Hawaii, and Puerto Rico). Medicare beneficiaries constitute more than 98% of the US population age 65 and older.⁵⁹ NHATS used a stratified three-stage sample design: 95 individual counties or groups of counties were selected as primary sampling units (PSUs), and 655 zip codes were sampled from them as secondary sampling units (SSUs). Individuals were sampled from the Medicare enrollment data bases (EDBs) of the SSUs with oversampling of non-Hispanic black individuals and the oldest old. The probabilities of sampling at all three stages aimed to create equal probability samples and achieve the goals of sample sizes by age group and race/ethnicity. The waves 5 to 8 were selected because the measures of the physical environment were included since wave 4, and the NHATS sample was replenished in the 5th wave (2015). Starting the analysis from wave 5 prevents the problem of having a high percentage of missing in the baseline of longitudinal analysis. In total, there were 8,334 complete cases in wave 5 data collection.⁶⁰ The overall response rate at wave 5 was 76%, which consist of 96% of the continuing sample and 63% of the newly enrolled sample.⁶⁰ Table 1 presents the baseline (wave 5) NHATS sample composition by race/ethnicity and age groups.

Table 1. wave 5 NHATS sample characteristics

Demographic variables	n	%
Age groups		
65-74	2847	34.16
75-84	3124	37.48
85+	2067	24.8
Gender-female	4927	59.12
Race		
Non-Hispanic White	5703	58.43
Non-Hispanic Black	1710	20.52
Others	244	2.93
Hispanic	467	5.6
Social Isolation		
least isolation	3095	37.96
somewhat isolated	4084	50.08
Socially isolated	976	11.97
Cognition		
No dementia	5825	70.63
Possible dementia	779	9.45
Probably dementia	963	11.68

2.1.1a. Measurements

Physical environments were reported by the interviewers who visited the respondents at their home. **Standing in front of the respondent's home/building and looking around**, the interviewer rated on a four-point scale (1=none, 4=a lot) regarding 1) the amount of litter on the street, 2) graffiti on buildings and walls, 3) vacant or deserted houses or stores. The physical condition of the **respondent home** was rated dichotomously (yes/no), in terms of, 1) any broken windows, 2) a crumbling foundation or open holes, 3) missing bricks or siding, 4) roof problems, 5) uneven walking surface or broken steps, 6) continuous sidewalks in both directions (reverse coded). The physical condition **inside the responder's home/apartment** was assessed by observing the existence of 1) peeling paint, 2) evidence of pests, 3) broken furniture, 4) flooring need of repair, 5) tripping hazards, and 6) cluttered room. The scale score of the physical conditions of the community, home/building, and inside home/apartments will be derived from the sum score of the corresponding items. The use of interviewers' observation could provide more accurate information regarding the physical environment of each participant than using the census data by zip code, which tends to reduce the complexity of the real-life condition.

Social environments. Respondents reported whether the people in the community know each other very well, willing to help each other, and people in the community can be trusted on a three-point scale (1=agree a lot, 2=agree a little, 3=do not agree). A sum score of the three items will be used as the scale score for the social environment.

ICT use. NHATS assessed three types of ICT use: instrumental internet use, social internet use, and internet use for medical purposes. **Instrumental** internet was measured by asking the respondent's weather they shop online and pay bills or banking with the internet. **Social** use was measured with one item asking in the last month if the respondent went online to visit social network sites (such as Facebook or LinkedIn). Use the internet for **medical** purposes had four items: order or refill prescriptions, contact medical providers, handle health insurance matters, and get information about one's health conditions. The three aspects of ICT use will be used as moderators that are independent of each other.

Social isolation was constructed with six dichotomously coded indicators: no marriage or partnership, no family identified to talk with about important things, no friend identified to talk with about important things, never meeting in person with family or friends in the past month, no participation in religious service, and no club, class or organization activities participation.³³ All items of social isolation will be coded as a higher indicates a more isolated situation. The scale score is the sum of six dichotomous indicators, ranging from 0-6. Scores less than 1 represent individuals who were least isolation; scale score 2-3 represent who was somewhat isolated, and scores equal or above 4 indicates social isolation.

Cognitive Health. For sampled persons self-responded cases, NHATS study assessed 1) **memory**: using immediate and delayed words recall; 2) **orientation**: date, month, year and day of the week, naming the President and Vice President of the United States; and 3) **executive function** measured by clock drawing test.⁶¹ If the interview were conducted with a proxy, AD-8 Dementia Screening Interview was used to assess memory, temporal orientation, judgment, and function.^{62,63}

Control variables. Demographics including age, gender, race, education level, marital status, income, and self-rated health have been reported in existing literature as predictors for social isolation among older adults^{13,20,64}. Therefore, the current study includes the variables listed above as control variables.

2.1.1b. Quantitative data analysis strategies

Data management and variable re-coding will be conducted using Stata 15 SE. Prior to model building, univariate analysis and correlation of key variables will be run to estimate the effect size and detect any potential multicollinearity issues. Latent growth curve modeling (LGCM) will be employed to model the longitudinal relationships between physical environments, social environments, social isolation, and cognitive health. LGCM allows nested models at hierarchical levels, such as observations of the same individual are nested within the person, individuals live in the same neighborhood would have similar physical environments with regard to disorder at the community level. As a special case of structural equation modeling (SEM), LGCM is well suited for modeling mediation paths (H3, H6). Besides, the LGCM is capable of testing the moderating effects of ICT use (H7).⁶⁵ Root mean square error of approximation (RMSEA), comparative fit index (CFI), and Tucker-Lewis index (TLI) were used to evaluate the extent to which each model fits the data. RMSWEA<0.05, CFI, and TFI > 0.90 would be considered as an acceptable model fit; CFI and TFI greater than 0.95 indicate good model fit.⁶⁶ LGCM models will be run with Mplus 8. Full Information Maximum Likelihood (FIML) estimation was used to handle missing data in models. FIML is a single step maximum likelihood approach that is widely used in structural equation modeling.^{67,68} Sampling weight will be applied to the models to match the representativeness of the racial/ethnic groups as designed in the sampling strategies.

2.1.2. Qualitative inquiry with I-CONNECT participants

2.1.2a. Data collection and procedure

Focus group participants will be recruited purposefully among the current I-CONNECT participants. Based on the strategy of maximum variation,⁶⁹ individuals with diverse living environment conditions will be selected to participate in the focus group discussion. Focus group members will be selected from both study sites of I-CONNECT (OHSU & UM). Two focus groups will be held at each site: one with intervention group participants, the other with control group participants. It is targeted for each focus group to have 8 participants (total N=32). To avoid contamination of the intervention effect, only participants who have completed the I-CONNECT study will be invited to join the focus group study. Given the study timeline of the I-CONNECT, the qualitative inquiry will extend from F99 phase to K00 phase. The interview guide will be developed informed by concepts in ETA and in consultation with experts with qualitative/mixed-methods research specialties (Palinkas) and with the best knowledge of I-CONNECT participants (Dodge). The PD/PI (Yu) will facilitate the focus group discussion at both OHSU and UM.

2.1.2b. Qualitative data analysis

Conversations in focus group sessions will be audio-recorded and then transcribe verbatim for analysis. The transcripts will be analyzed using inductive content analysis method by at least two trained independent coders.⁷⁰ The coders are going to collaborate with qualitative analysis software Nvivo 12. The qualitative data coding team will meet regularly to discuss the coding framework, procedure, and to reconcile different opinions and interpretations of the quotations in the transcripts. The team will employ Cohen's kappa coefficient as the indicator for inter-rater reliability.⁷¹ Themes identified in the coding process will be reorganized and made sense of according to the personal and living environmental characteristics of the participants.

2.1.3. Integration of quantitative and qualitative data

The qualitative and quantitative inquires of the dissertation work will occur concurrently. The main purpose of adopting the mixed-methods is using qualitative data to explain why there might be effects of physical and social environments on social isolation. The qualitative focus groups expanded the discussions on the findings identified in the quantitative analysis.⁷² Meanwhile, participant's insights on designing a sustainable environmental-focused technology program to address social isolation and promote cognitive health will be elicited. The findings of both qualitative and quantitative components of the dissertation work together as foundations for developing an intervention program that can be pilot tested in the postdoctoral training phase and beyond.

2.2. K00- postdoctoral stage

2.2.1. I-CONNECT

2.2.1a. Study design

I-CONNECT study is being conducted in collaboration by Oregon Health and Science University (OHSU) and the University of Michigan (UM) (ClinicalTrials.gov Identifier: NCT02871921). One hundred sixty socially isolated adults aged 75 plus will be recruited at each site (OHSU & UM) and randomized into either the video-chat intervention group and the control group. As of October 2019, 105 subjects are already participating in the trial.

The inclusion criteria are aged 75 or older, consent to magnetic resonance imaging (MRI) if eligible, socially isolated, adequate vision and hearing, understand English, and normal cognition or MCI (based on consensus diagnosis). The intervention group participants received computers and internet service during the intervention, which ensures the participants can video chat with study staff for 30 minutes/day, 4 times/week for 6 months (high dose), and 2 times/week for an additional 6 months (maintenance dose). Study staff briefly check-in by phone call (approximately 10 minutes) with both intervention and control group participants once per week. In-home neuropsychological assessments will be administered at baseline, 6 months, and 12 months. All participants would have MRI scans at baseline and 6 months if MRI were safe for them. All video chat and in-home assessment sessions are being recorded for language analysis. Participants at OHSU will have their medication compliance tracked using electronic pillboxes. The estimated primary completion date for I-CONNECT study is Oct 2022. The expected study completion date, including the data analysis phase, is Oct 2023.

2.2.1b. Measurements

Cognitive functioning. The primary intervention outcomes are executive and memory functioning assessed with cognitive tests using National Alzheimer's Coordinating Center Uniform Data Set neuropsychological battery (UDS V3: https://www.alz.washington.edu/WEB/dataforms_main.html), cross-validated by NIH Toolbox.⁷³ Region-specific structural and functional pre-post changes in the brain are evaluated with magnetic resonance imaging (MRI). **Speech and language** characteristics are captured during each video chats as well as in-home assessments. **Medical adherence** measured with electronic pillboxes. Compared to traditional self-reported medical adherence, the digital biomarker collected with electronic pillboxes can eliminate the social desirability bias and record the exact time when the participants opened the pillboxes.⁷⁴ **Social Isolation** was defined as: i. Score ≤ 12 on the 6-item Lubben Social Network Scale (LSNS-6);⁷⁵ or ii. Engages in conversations lasting 30 minutes or longer no more than twice per week, per subject self-report.

2.2.1c. Intervention effects evaluation plan

Among the outcomes measured in I-CONNECT, my interests focus on cognitive health and social isolation. I will collaborate with I-CONNECT team to examine the intervention effects. The high dose (baseline to 6 months) and the maintenance dose (6 months to 12 months) will be examined separately. Data will be analyzed to evaluate the differences in cognitive function change and extent of social isolation from baseline to 6 months between the intervention and the control groups. Similar analyses will be conducted for the change between 6 months to 12 months. Finally, the combined intervention effects of the high dose and the maintenance dose intervention on cognition and social isolation will be evaluated.

2.2.2. Co-design a technology-facilitated community-based intervention with older adult participants

Despite the empirically supported benefits of ICT for older adults' wellbeing, the majority of the currently available technology tools, such as social media sites, mobile health apps, are not designed with older adults' unique needs in mind. Older people often are not included as part of the intervention/technological tool development team. It was not until recent years the researchers in the health service and related fields start to employ a co-design approach to try to develop programs from the perspective of the populations it aims to serve.⁷⁶⁻⁷⁸ On the other hand, anxiety and resistance remains among the older adult population when it comes to learn, trust and safely adopt new technologies.⁷⁹⁻⁸¹ I aspire to co-design a technology-facilitated community intervention to reduce social isolation and to prevent cognitive decline at the K00 research phase, which can lead to future research agenda on pilot testing the potential effects and feasibility of the intervention program after I become a tenure-track faculty.

Previous research summarized six steps in developing a quality intervention, which are 1) defining and analyzing the problem; 2) identifying modifiable factors; 3) deciding on the intervention mechanism; 4) clarify the intervention delivery process; 5) testing and revising the intervention; 6) a more rigorous effectiveness evaluation.⁸² The proposed dissertation work will address steps 1 to 4 in intervention development. More specifically, the longitudinal quantitative analysis will analyze the problem of social isolation and cognitive decline, identifying physical and social environmental factors as modifiable points and potential intervention mechanisms. The focus groups with I-CONNECT participants will help to collect preliminary information on the participant's opinion towards the intervention design and clarify the intervention delivery process that fits the needs of the older adults.

In the K00 phase, I plan to synthesize the evidence build with I-CONNECT intervention and conduct more qualitative co-design sessions with isolated older adults who are at risk of cognitive decline. Collaborating with I-CONNECT team, my research activities at the postdoctoral stage will further clarify the intervention mechanism and intervention delivery process. Experts panels will be formulated to consult the development of the intervention, which includes the sponsors of the current proposal and other researchers and experts I will meet with in the future.

Respective Contributions

This research and training proposal was informed by the applicant's previous research, clinical experience, coursework, and theoretical thinking toward the research questions. The research and training plans were developed, reviewed, and edited in collaboration with all sponsors and cosponsors. Preparing for this application, I had weekly consultation meetings with Drs. Chi and Wu, who were involved in the conceptualization of the specific aims, selection of theoretical framework, development of training plan, and providing feedback to many drafts of the proposal. As the PI of the parent study for the postdoctoral research, Dr. Dodge worked with me to formulate the postdoctoral research and training plan. She oriented me with resources and personals available at OHSU and OADC. Dr. Dodge helped me develop the specific aim and reviewed and provided feedback on several drafts of the proposal. Dr. Palinkas guided the planning of the mixed-methods research design, helped to develop the training plan, and provided invaluable input on the proposal drafts.

Selection of Sponsor & Institution

Dr. Iris Chi is a full professor at USC SSW. Dr. Chi has over 30 years of research experience in gerontology and social work. She has published widely on the health and mental health issues among older Chinese adults. Dr. Chi has expertise in survey research, secondary data analysis, and causal inference with longitudinal data. She has mentored Ms. Yu for the past three years and served as a co-chair for Ms. Yu's qualifying exam. Dr. Chi is selected as the primary sponsor of this application because of her extensive experience in conducting research with the older adult population. Dr. Chi has been working closely with community-based organizations to conduct research and implement interventions. She will be able to guide Ms. Yu to develop professionally as an intervention scientist focus on community-based research. Dr. Chi has helped Ms. Yu to formulate a clear conceptualization of her research questions and training agenda. Dr. Chi will continue to contribute to Ms. Yu's professional development through weekly meetings, providing guidance in conducting responsible research, theoretical interpretations of empirical findings, and intervention development and adaptation.

Dr. Shinyi Wu is an associate professor of Social Work and Industrial & System Engineering at USC. She employs an engineering approach to study behavioral health. Her research has focused on improving system efficiency in delivering health care and reducing health disparities. Dr. Wu has served as Ms. Yu's co-mentor and the co-chair of her qualifying exam. In the past three years of collaboration, Dr. Wu has worked with Ms. Yu to examine the feasibility and potential of harnessing the capacity of mobile technology to promote self-management of chronic disease among older type 2 diabetes patients. Dr. Wu is selected as a co-sponsor because of her invaluable expertise in employing technology as an intervention tool for the older adult population. She also leads a pilot study on understanding the challenges of coordinating care for Latinos with AD. In the F99 phase, Dr. Wu will directly work with Ms. Yu to continue to examine the effects of ICT use on social well-being and cognitive health. After Ms. Yu graduates from USC and starts the K00 phase of research, Dr. Wu will consult with Ms. Yu on technology-related research topics.

Dr. Hiroko Dodge is the Kevreson Professor of Neurology at the University of Michigan, with a joint appointment as Professor of Neurology at Oregon Health and Science University. She has been successfully directing two Data Management and Biostatistics Core over the past 10 years, one at Michigan ADC and another at Layton Aging and Alzheimer's Disease Center at OHSU. She has previously mentored 17 PhD and postdoctoral fellows, and four career development awardees (K01/K23/K25 mechanisms). Dr. Dodge and Ms. Yu have established mentoring relationships through the development of this F99/K00 application. For proposed project under both F99 and K00, Dr. Dodge will be co-sponsor, providing training in the development of clinical trial designs, recruitment of underserved population (e.g., socially isolated oldest-old at high risk of developing dementia), selection of clinical trial outcomes, clinical trial enrichment and analyses of high-frequency digital biomarker longitudinal data. She will also provide assistance to Ms. Yu for familiarizing herself with dementia research, especially in the area of dementia and social isolation.

Dr. Lawrence Palinkas is a full professor of Social Work, Anthropology, and Preventative Medicine at USC. He has received continued support from NIDA, NIMH, and NICHD. Dr. Palinkas has published broadly on innovative mixed-methods and qualitative research design that is used to address the health needs of older adults. His methodological expertise in mixed-methods research will help to ensure the rigor in design, data collection, and analysis of the mixed-methods component of the proposed dissertation study. Dr. Palinkas's recent research focuses on the implementation process of evidence-based interventions. He will guide Ms. Yu to conduct quality mixed-methods research that could inform the development of a community-based intervention for social isolation and cognitive health. Dr. Palinkas will have monthly consultation meetings with Ms. Yu.

University of Southern California is the primary training site for Ms. Yu's F99 phase of dissertation research. Oregon Health and Science University will be the prospect training sites for the postdoctoral research and training at the K00 phase. USC SSW is a prominent academic and research university that has been training leaders in diverse study fields. The resources at USC will fully support Ms. Yu's training goals. Ms. Yu is a member of the USC Edward R. Roybal Institute on Aging and has access to the frontier aging research. Research activities at the Roybal institute address complex issues in aging with an interdisciplinary approach. USC SSW also provides the applicant with opportunities to pursue training opportunities at the university level, including coursework at the Keck School of Medicine and Davis School of gerontology, which laid the foundations for Ms. Yu to study cognitive health and social well-being with the older adult population. The OHSU School of Neurology has a special emphasis on using technology-facilitated approach to address the problem of cognitive decline in older people. Ms. Yu's research interests match well with the research emphasis at OHSU. OHSU will support the applicant's research and training agenda in the postdoctoral stage.

Training in Responsible Conduct of Research

The USC Suzanne Dworak-Peck School of Social Work (USC SSW) provides instruction for the responsible conduct of research in several venues: (a) research methods courses; (b) professoriate workshops and scholarly lectures; (c) the Hamovitch Center for Science in the Human Services; and (d) IRB research training courses.

Research methods courses. Format: Courses are designed for small in-person group discussions (7 or 8 students per course). Subject matter: Courses cover a wide range of topics, including ethical research conduct, conflict of interest, protecting human subjects, community-based and collaborative research practices, peer review, publication ethics and authorship, responsibilities to humanity, avoiding and handling misconduct, and ethical and unethical study designs. Faculty participation: Tenure-track faculty members instruct all courses. Duration of instruction: Courses meet weekly for 3 to 4 hours for the duration of the semester. Frequency of instruction: The PI successfully completed three research methods courses in fall 2017, spring 2018, and fall 2018 (see biosketch for reference).

Professional development workshops and scholarly lectures. Format: Workshops and lectures are designed for face-to-face discussion among doctoral students. Subject matter: Professional development workshops and scholarly lectures cover topics relevant to becoming a successful researcher: responsible and ethical conduct in publishing, authorship, peer reviews, data collection and analysis, study design, grant proposals, use of financial awards, and managing collaborative research teams. Faculty participation: Senior and tenure-track research doctoral faculty members, in partnership with research professionals at the Hamovitch Center, conduct the workshops and lectures. Duration of instruction: Workshops and lectures range from 1 to 2 hours and are scheduled monthly throughout the academic year. Attendance is mandatory for all doctoral students. Frequency of instruction: The PI has successfully completed 17 professional development workshops and scholarly lectures (2017 to present) and plans to continue to attend the workshops and scholarly lectures.

Hamovitch Center for Science in the Human Services. (see Facilities and Other Resources document for more information). Format: scholarly workshops, lectures, and individual consultation. Subject matter: The Hamovitch Center specializes in all aspects of conducting research and provides workshops, lectures, and individual consultation from full-time professionals with expertise in conducting responsible and quality research. Faculty participation: Full-time senior research faculty members and research professionals are available for USC faculty members and doctoral students engaged in research. Duration of instruction: Workshops and lectures are scheduled semiquarterly throughout the academic year and individual consultation is available as needed. Frequency of instruction: The PI has a working relationship with several full-time professionals at the Hamovitch Center, engaging in activities that include data analysis consultation, manuscript writing, study design, and grant proposals.

IRB courses. Format: Online training courses. Subject matter: Courses completed are (December 2016): Human Subjects; Good clinical practice; Conflicts of Interest, and Health Insurance Portability and Accountability Act (HIPAA) Privacy Education. Faculty participation: All sponsor/cosponsors have completed these trainings. Duration of instruction: These IRB course trainings are required for all proposed studies at USC, as are periodic refresher courses. Frequency of instruction: The PI is scheduled for refresher Human Subjects and Good clinical practice courses in December 2019.

Besides the trainings made available at USC SSW, the PI has engaged in the Digital Scholar Training Program provided by USC Clinical and Translational Science Institute (CTSI), where a series of responsible conduct of research topics are discussed.

CTSI Digital Scholar Training Program. Format: Online training courses. Subject matter: This program helps researchers who are involved in the health sciences to develop competency in digital research practices. Subjects covered including selecting right journal for publication, collaboration with industry, and participant recruitment using non-traditional approaches etc. Faculty participation: tenured faculty members and senior scientists are invited to give webinar presentations. Duration of instruction: Online courses typically last for an hour with Q&A sessions at the end. Frequency of instruction: The CTSI digital scholar training program schedules online courses every month. The PI has signed up for the notifications for webinars. She has attended about 5 webinars and plan to continue to attend trainings that suit her learning needs.

Sponsor and Co-Sponsor Statements

A. Research Support Available

Chi	
Wu	
Dodge	
NIA R01 AG051628	Conversational Engagement as a Means to Delay Alzheimer's Disease Onset: Phase II (Dodge, PI), 09/01/16 – 05/31/21
NIA R01 AG056102	Web-enabled social interaction to delay cognitive decline among seniors with MCI: Phase I (Dodge, PI), 07/01/17 – 03/31/22
NIA U2C AG054397	ORCATECH Collaborative Aging (in Place) Research Using Technology (CART) (Kaye, PI), 09/01/16 - 08/31/20
Palinkas	
NIDA P30 DA027828	Center for Prevention Implementation Methods for Drug Abuse and Sex Risk Behavior (Brown, PI), 9/1/16 –6/30/21
NIMH P50 MH113662-01A1	Accelerator Strategies for States to Improve System Transformations Affecting Children, Youth and Families (Hoagwood, PI), 9/21/19 – 4/30/23
NICHD R01 HD092489-01A1	Dissemination and implementation of the Safe Environment for Every Kid (SEEK) Model for Preventing Child Maltreatment. (Dubowitz, PI), 4/1/19 – 8/31/23

B. Sponsor's/co-sponsor's Previous Fellows/Trainees

Chi – has mentored 32 doctoral students and postdoc fellows in total

Xinming Song	1991-1995	Professor, Peking University
Vivian Lou	1993-1999	Associate Professor, University of Hong Kong
Angela Leung	2000-2007	Associate Professor, Hong Kong Polytechnic University
Man Guo	2005-2011	Associate Professor, University of Iowa
Ling Xu	2007-2012	Assistant Professor, University of Texas, Arlington

Wu – has mentored 15 doctoral students and postdoc fellows in total

Irene Vidyanti	2009-2014	Data Scientist, Los Angeles County Chief Information Office
Magaly Ramirez	2011-2016	Assistant Professor, University of Washington
Haomiao Jin	2012-2016	Research Assistant Professor, University of Southern California

Dodge – has mentored 17 doctoral students and postdoc fellows in total

Adriana Seeley	2014-2017	Assistant Professor, Minnesota VA hospital
Meysam Asgari	2014-2016	Assistant Professor, Oregon Health and Science University
Junko Nishihira	2014-2016	Assistant Professor, Ryukyu University
Jiayu Zhou	2016-2018	Assistant Professor, Michigan State University
Ming Li	2017-2018	Researcher, Alibaba Seattle

Palinkas – has mentored 70 doctoral students and postdoc fellows in total

Maxwell Davis	2007-2008	Director of the CALSWEC Mental Health Program, University of California, Berkeley
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Ian Holloway	2010–2012	Associate Professor, University of California, Los Angeles
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Kim Brimhall	2016-2017	Assistant Professor, Binghamton University
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C. Training Plan, Environment, Research Facilities

Ms. Yu proposes to examine the longitudinal influence of physical and social living environments on social isolation and cognitive health among older adults with National Health & Aging Trends (NHATS) data. The proposed dissertation will be the known first to contextualize the role of ICT use through examining the interaction effects of the physical environment with ICT use and the social environment with ICT use on social isolation and cognitive decline. An ecological perspective has rarely been adopted to studying the intertwined relationships between social well-being and cognitive health. With the population aging rate continues to increase, Ms. Yu's research will enhance the research field's understanding of the ecological support system on mental health and cognition in the rapidly changing society. Ms. Yu proposes to develop and pilot testing a community-based intervention that has the potential to provide a sustainable solution to keep the older adults socially connected and prevent the onset of dementia. Her work will be meaningful to address the challenge of social isolation and the burden of care brought by the increasingly prevalent ADRD conditions. Ms. Yu's training plan incorporates both coursework, seminars, and mentorship opportunities provided by sponsors with diverse and complementary skillsets. The proposed research and training agenda works together to advance Ms. Yu's research skills and scholarship.

With the receipt of this award, Ms. Yu will have the opportunity to receive training from an outstanding mentoring team made up of leaders in the fields of Neurology, Engineering, Anthropology, and Social Work. Each of the sponsors has influential research works committed to improving the quality of lives for the older adult population. The sponsor team members will collaborate to aid Ms. Yu to achieve her long term career goal of becoming an independent social scientist and an impactful intervention researcher that dedicated to improve social well-being and prevent cognitive decline. Ms. Yu will work with Dr. Wu to further gain expertise in using technology-facilitated tools to improve health outcomes and increase access to health care for the populations in need. She will work with Dr. Dodge to increase her understanding of cognitive health and dementia research, to grow her capacity of neuroimaging methodology and using digital biomarkers in her research. Ms. Yu will work with Dr. Palinkas to refine her skills in mixed-methods research and receive consultation on designing and implementing community-based health interventions. Ms. Yu will apply the skills developed from the individualized training plan to write her dissertation, seek to publish in high-impact journals, and develop a research agenda beyond the postdoctoral training stage than leads to her career success.

I serve as the primary sponsor of Ms. Yu's F99/K00 application. I will be the coordinator for communication among the co-sponsors. Depending on my 30 years of experience in conducting research on health and mental health outcomes with the older adult population, I will provide content and professional development expertise on: (a) interpretation of the longitudinal analysis and qualitative results; (b) integration of qualitative and quantitative findings; (c) developing and/or adapting social isolation and cognitive health intervention in collaboration with the target age group and community-based organizations; and (d) scientific writing and navigating the peer-review process. Working with the sponsoring team, Ms. Yu has identified the following training goals and listed specific plans that lead to achieving these goals.

Training Goal 1: Ms. Yu will increase knowledge in prevention, intervention, and current practice and policy pertain cognitive decline in normal aging and the development of ADRD. Dr. Wu is currently conducting research with older adults with ADRD and their family caregivers. She will connect Ms. Yu with researchers with similar interests at the USC Alzheimer's Disease Research Center (ADRC). Ms. Yu will work with Dr. Wu and myself to develop a literature review database on cognitive health and ADRD research. Dr. Wu leads a health system journal club with graduate students under her supervision. At the bi-weekly journal club, Ms. Yu will have opportunities to present her learned ADRD related knowledge. She will have interactive discussions and receive feedback from peers and Dr. Wu and myself. The journal club and involvement at the USC ADRC will formulate an intellectual community and academic support system that will benefit Ms. Yu's professional development and assist her in achieving the training goal 1.

Dr. Dodge is an expert in pharmacological and non-pharmacological intervention strategies are urgently needed to delay the onset of dementia and reduce its societal burden. She has contributed to the knowledge of distinguishing the normal cognitive aging and pathological cognitive decline. Dr. Dodge is the Principal Investigator of Internet-based Conversational Engagement Clinical Trial (I-CONNECT, NIA R01AG056102,

R01AG056628 & R56AG051628, ClinicalTrials.gov Identifier: NCT02871921), examining the effect of technology-facilitated conversation engagement on delaying cognitive decline. She is the ideal senior scientist in Neurology to work with Ms. Yu to develop her knowledge base in cognitive health and AD/DR through hands-on research experiences and directed learning. Dr. Dodge is currently collaborating with Ms. Yu to prepare to apply for a pilot grant at OHSU to conduct qualitative focus groups to understand the older adults' perspective on the association between living environments, social engagement, and cognitive health. At the F99 phase, Ms. Yu will meet with Dr. Dodge using online video conferencing tools at least once in a month. Ms. Yu will also communicate with Dr. Dodge with the opportunities provided by conducting a qualitative pilot study and collaborating on manuscripts. Ms. Yu will have access to Dr. Dodge with any substantial and methodological questions relevant to AD/DR prevention, intervention, and policy research. As the director of the data core at OADC, Dr. Dodge will connect Ms. Yu with scholars at Layton Aging and Alzheimer's Disease Center at Oregon Health and Science University (OADC). Dr. Dodge will direct Ms. Yu to other research cores at OADC as her learning and research need requests. Ms. Yu will attend the monthly OADC investigator meetings using Zoom at her F99 phase, and in-person at the K00 phase. At these investigator meetings, Ms. Yu will attend the research grant presentation by leading researchers in AD/DR related domains.

Ms. Yu has selected courses that suit her learning needs. In the first year of F99, She will take a cognitive decline and AD/DR special topic course at USC Leonard Davis School of Gerontology. The course is designed to cover the normal and abnormal brain and cognitive function changes in the aging process. Ms. Yu will learn from the course approaches of prevention, early detection, and management of AD/DR. Ms. Yu will expand her academic network by submitting abstracts and presenting in cognitive health-related conferences, such as the Alzheimer's Association International Conference (AAIC), Gerontological Association of America annual meeting (GSA), and Clinical Trials in Alzheimer's Disease Conference (CTAD).

Training Goal 2: Ms. Yu will gain skills in analyzing high-frequency digital biomarkers longitudinal data and employing neuroimaging methods. Ms. Yu proposes to use digital biomarkers and neuroimaging outcomes in her research at the K00 phase. Using IMTOP data, Ms. Yu has first-authored a manuscript using blood test results, such as A1C, blood pressure, and cholesterol, to evaluate the mhealth intervention on type 2 diabetes clinical outcomes. However, Ms. Yu does not have experience in using biomarker outcomes in study projects with social isolation and cognitive health-related research. She will receive bi-weekly directed readings on the example studies using biomarkers and neuro-imaging to study changes in cognition functioning. Ms. Yu will present her learning outcomes and consult with Dr. Dodge any specific methodological questions she might have in the regular online meeting. Dr. Wu and I will also regularly with Ms. Yu to get updates on her learning of using biomarkers and neuroimaging in her research. The OADC is equipped with a neuroimaging research core and a biomarker research core. In the K00 training phase, Ms. Yu will attend training and seminars hosted by the two research cores. She will also be able to work on the I-CONNECT data to gain hands-on experience in neuroimaging and biomarker methodologies.

Dr. Dodge has a strong background in statistical methods. She will supervise Ms. Yu's self-directed learning of using R programming to manage longitudinal digital bio-marker data collected in high frequency. Dr. Dodge will also connect Ms. Yu with other researchers at OADC with specialties in neuroimaging methods and analyzing digital biomarkers as clinical intervention outcomes.

Training Goal 3: Ms. Yu will receive advanced training on design, implement, and evaluate community-based interventions. Ms. Yu has a clear vision of reducing social isolation and delay the onset of dementia with community-based intervention research. I have built long-term collaborative relationships with community-based health and social service providers in southern California. Ms. Yu has learned how to build collaborative relationships with community partners, participated in recruitment, and data collection by serving as a co-investigator in the pilot study effort employing a co-design approach to develop a caregiver self-management program (CSMP) app for Chinese caregivers. She has further advanced her experience in community-based research by working with Dr. Wu and myself on pilot studies examining the acceptability and feasibility of IMTOP intervention among recent immigrants in the United States and for low-income older residents of affordable retirement communities. With my experiences in intervention and behavioral science research, I will continue to guide Ms. Yu to conceptualize her intervention study and provide constructive feedback to her theoretical reasoning and design of the intervention. Dr. Wu and I will discuss with Ms. Yu in our weekly mentoring meetings the practical, intervention, and policy interventions for her findings of the longitudinal analysis using NHATS data. Dr. Wu has served as Principle Investigator for several intervention studies examining the effectiveness of employing mobile technology as enabling tools for improving the health outcomes of older adults. She is uniquely qualified to guide Kexin to make sense of the analytical results of survey data related to ICT use in the older adult population and help her to translate the empirical findings to intervention mechanisms. I will guide Kexin to

interpret the hypothesized longitudinal association between physical and social environments and social isolation with my extensive research experiences on researching with the older adult population and their support networks.

Dr. Palinkas is a leader in the mixed-methods research methodology. His seminal paper on the new development in purposeful sampling for mixed-methods implementation research has been cited for over 2,500 times. Ms. Yu proposes to adopt a mixed-methods approach for her dissertation to build knowledge with both longitudinal survey data and with qualitative perspectives and insights from socially isolated older adults recruited from the I-CONNECT parent intervention study. Dr. Palinkas is the ideal mentor to work with Ms. Yu to formulate a purposeful sampling, data collection, and analysis plan and work with Ms. Yu to prepare manuscripts reporting mixed-methods study findings. Dr. Palinkas is also an expert in implementation and dissemination science. Ms. Yu will consult Dr. Palinkas on the process of intervention implementation and develop a draft of the intervention research proposal with his guidance. Dr. Palinkas will have monthly consultation meetings with Ms. Yu. She will be able to receive individualized training on methodological and substantive topics.

With the recipient of the award, Ms. Yu will have the opportunity to receive immersion training with I-CONNECT fieldwork, observe how the team monitors recruitment, enrollment, and randomization in the K00 phase. She will take an online course provided by UCSF on study design for intervention research in real-world settings. The course fits Ms. Yu's training goal particularly well, given her interests in conducting community-based intervention research from the social ecology perspective. Ms. Yu will also apply for the National Institute of Neurological Disorders and Stroke (NINDS) supported clinical trial methodology course (R25NS088248) to receive two years of continuous training with both online and in-person formats. This course is designed to help junior investigators develop scientifically rigorous, yet practical clinical trial protocols, and to focus funding mechanisms as a critical trial planning activity. Taking this course will aid Ms. Yu to grow both scholarly in knowledge and professional in grant preparation.

Dr. Wu, Dr. Dodge, Dr. Palinkas, and I formulate an experienced and committed sponsor/co-sponsor team to contribute to Ms. Yu's professional development and facilitate her to achieve her long-term career goal of becoming an independent scientist and impactful intervention researcher. As the primary sponsor of this F99/K00 proposal, I will be the coordinator for the communication between co-sponsors. Ms. Yu will take the initiative to request a meeting with all sponsors and co-sponsors in the process of conducting the proposed research activities. All four sponsors and co-sponsors will meet in-person once in a year at major scientific meetings, such as the GSA annual conference.

Environment and Research Facilities

Ms. Yu has been a student of Dr. Palinkas, a mentee of Dr. Wu and myself for the past three years. Ms. Yu has established a collaborative relationship with Dr. Dodge through the development of this F99/K00 application. This recently established mentoring relationship has already been proven extremely valuable to Ms. Yu. Dr. Dodge has been very instrumental in helping Ms. Yu refining her thinking in methodology and training plans related to her research at both dissertation and postdoctoral stages.

The Suzanne Dworak Peck School of Social Work is among the top schools of social work in the nation according to the U.S. News & World Report and has a vast amount of available resources. Ms. Yu also has access to the school's research center, the Hamovitch Center for Science in the Human Services; the USC Edward R. Roybal Institute on Aging; the USC ADRC. Details regarding these resources are thoroughly described in the Facilities and Other Resources and Description of Institutional Environment and Commitment to Training sections of the proposal.

D. Number of Fellows/Trainees to be Supervised During the Fellowship

Besides working with Ms. Yu, I will be supervising Mandong Liu, co-mentoring Ruotong Liu, and Yi-Hsien Tung with Dr. Wu during the period of the award. Dr. Wu will also be independently supervising Anthony Nguyen, a PhD student in Industrial and Systemic Engineering. Because Dr. Wu and myself work closely on research projects and share responsibilities of mentoring PhD students, we will have sufficient time and resources to support Ms. Yu's dissertation and postdoctoral research. Dr. Dodge will be mentoring three other postdoctoral fellows, Patrick Pruitt, Chao-Yi Wu, and WanTai (Michael) AuYeung, during the period of the F99/K00 award. Dr. Palinkas will be supervising Abigail Palmer-Molina, a recent recipient of F31.

E. Applicant's qualifications and Potential for a Research Career

Before joining the MSW/PhD dual-degree program at USC SSW, Ms. Yu had two years of experience working as a research assistant in the cognition and aging lab under the supervision of Dr. Shulin Chen. Ms. Yu received excellent research training in mental health and developed strong interests in aging research. She was selected

to work as a research intern at the University of Alberta, working on the Translational Research in Elder Care program under the supervision of Drs. Carole Estabrooks and Matthias Hoben. Ms. Yu worked with leading researchers in Canada and gained hands-on research experience studying the effects of working environments of nursing homes on staffs' mental and physical health. This valuable training experience resulted in a systematic review focused on barriers and facilitators of oral health care for older people published in the International journal of nursing studies.

Since she joined the MSW/ PhD program at USC, Ms. Yu has been an integral part of our research team and has developed to become an emerging scholar in the scientific community. Ms. Yu's critical thinking, understanding of the literature, methodological skills are impressive. For the past three years of intensive training in the dual degree program, Ms. Yu not only gained her MSW degree but also has first-authored 12 conference presentations and developed high-quality manuscripts.

Ms. Yu has been a vital contributor to our research projects during the past three years. She has been actively engaged in grant proposal preparation, survey and qualitative interview guide design, and data collection with caregivers and older adults with disadvantaged backgrounds. Ms. Yu's first-authored work using the IMTOP study data was recognized as the winner of the 2018 APHA Nobuo Maeda International Research Award. Driven by her passion for employing technology to improve health outcomes, Ms. Yu led the IMTOP pilot study. Ms. Yu trained and supervised student workers to recruit participants at community-based organizations. The pilot study provides essential data for adapting the IMTOP intervention to the recent and first-generation immigrants with Type 2 diabetes in the greater Los Angeles. Ms. Yu worked with low-income retirement communities to evaluate the needs of a mHealth intervention within this specific living environment. This research experience sparked Ms. Yu's passion for studying the physical and social environments on social isolation and cognitive health as she observed the impacts of living environments on social and health outcomes.

Furthermore, Ms. Yu has been actively engaging in the co-design CSMP app pilot study with Chinese immigrant caregivers. She contributed significantly to the study design and data collective using semi-structured interviews. Ms. Yu is currently leading the qualitative data analysis and manuscript preparation for the co-design CSMP app study.

Besides her work with me, Ms. Yu has pursued her passion in addressing social isolation among older people using large scale, nationally representative longitudinal datasets, such as Health and Retirement Study and National Health & Aging Trends Study (NHATS). Her first-authored paper examining the longitudinal association between internet use and perceived loneliness in older adults is currently in revise and resubmission process. Ms. Yu has proposed her qualifying exam in May of 2019. She independently worked on the HRS longitudinal dataset and employed the cutting-edge random intercept cross-lagged modeling to detangle the sequence and directionality between perceived social isolation and accumulated geriatric conditions. The qualifying exam project bridges her research interests in physical health and social isolation. In the process of completing the proposed project, Ms. Yu demonstrated outstanding theoretical thinking, independence in learning, and advanced analytical skills. Ms. Yu passed her qualifying exam and advanced to candidacy in September of 2019. Ms. Yu is currently continuing to investigate the prevalence and predictors of geriatric conditions among homeless older adults in collaboration with Dr. Benjamin Henwood, an associate professor at USC SSW.

Ms. Yu is one of the most outstanding PhD students I have ever worked with during my over 30 years' experience of mentoring doctoral students. She is highly self-motivated and thrived on being challenged intellectually and challenging herself all the time. Ms. Yu is committed to improving the welfare of older adults through her research, especially for the socially isolated population who are at high risk of cognitive decline. Ms. Yu proactively seeks grant opportunities at a very early stage of her career, and she will be continuously working with an extraordinary sponsor team to refine her thinking, develop thoughtful and innovative research proposals, and secure support for her future independent research. I have every confidence that Ms. Yu is going to success as an impendent researcher.

USC Suzanne Dworak-Peck

School of Social Work

October 14, 2019

Dear Fellowship Selection Committee Members,

I am delighted to serve as Ms. Kexin Yu's primary sponsor for her F99/K00 proposal. I have known Kexin for three years in several capacities. Since 2016, together with Dr. Shinyi Wu I have been her co-mentor, co-chair of her qualifying examination and dissertation committees while she is pursuing her Ph.D. study at the University of Southern California. In addition, she works as my research assistant and participates in one of my research projects as a co-investigator. So far she has excelled in her doctoral training program and she is currently finalizing her dissertation proposal which is anticipated to be approved by her guidance committee by January of 2020. With her good academic track record, she is expected to submit her dissertation within fifteen months and graduate from our Ph.D. program in May of 2021.

As a doctoral student, Kexin has taken full advantage of our doctoral program by getting involved in several different research projects. She has worked closely with me and Dr. Wu on two of our mHealth studies designed to understand how technology can facilitate self-management of health among Type 2 diabetes patients and among immigrant caregivers who suffer from chronic conditions. Over the years using both our study and her own data, she has presented her work in several professional conferences such as the American Public Health Association (APHA), the Gerontological Society on Aging (GSA), and Society of Social Work Research (SSWR). She received the APHA Nobuo Maeda International Research Award in 2018. Kexin is the lead author for two manuscripts which are revised and resubmitted to two prestigious peer-review journals namely the Journal of Gerontology and the Diabetic Educator. Moreover, she co-authors in a number of our other manuscripts by making significant contributions. I believe these experiences provide a foundation for the unique, novel, and highly valuable research she proposes in her dissertation and onward.

I have watched Kexin grow in knowledge, sophistication, insight and confidence over the years. Her growth from hesitant young student to accomplished scholar has in fact been a wonder to behold. Driving this growth has been her truly extraordinary hard work and determination - thriving on being challenged and challenging herself all the time. She is one of the most impressive doctoral students whom I have had the privilege to mentor over my 30 years' experience in supervising doctoral students. My experience in mentoring her and collaborating with her on research gives me full confidence that Kexin will become a leader in her chosen field of research.

Kexin's commitment to the welfare of older Americans, strong research skills, and creativity are demonstrated in her dissertation proposal. She aims to examine the effect of living environments on social isolation by contextualizing the role of technology. Specifically, she plans to use mixed-methods approach which includes two longitudinal studies investigating the impact of the living environments on social

Suzanne Dworak-Peck School of Social Work, University of Southern California



isolation with 4 waves (i.e. wave 5 to 8) of data from the National Health & Aging Trend Study (NHATS) as well as a qualitative focus group study to explore the subjective perceptions of the environmental influence on older adults' social well-being. Conceptually, her proposed studies incorporate both the Ecological Theory of Aging and Information and Communication Technology to exemplify the multifaceted relationship among living environments, use of technology, and social isolation. The NHATS is uniquely suited for her research because questions regarding internal and external environmental conditions, social isolation as well as frequencies in technology use were asked over a 16 years' interval. The findings will add to our understanding of how living conditions influence social isolation and whether use of technology could intervene their relationships. The focus group discussion will provide insight on the complexity of environmental influence on technology use and social well-being. A distinctive strength of this study is using national representative longitudinal data which draws more rigorous conclusions regarding hypothesized casual effects in these relationships. Ultimately, her studies aims to provide information which will lead to a better understanding of ecological support systems and social well-being in a changing society. It is safe to say that social isolation among older adults will encounter even greater challenges in future given the continuing increase in aging population.

My research for more than 35 years has focused on health and mental health issues among older Chinese populations. I have the expertise in both quantitative, qualitative and mixed methods including ethnographic, focus group, survey research, longitudinal study, secondary data analysis, and intervention research on behavioral health studies. I have extensive knowledge on using interdisciplinary methods to study physical and psychosocial health issues among older adults. I am an honorary professor, fellow, consultant and advisor to more than 30 local and overseas universities and research institutes. I have a good track record of successful mentorship of doctoral students, post-doc fellows and tenure-track professors to become independent scholars both in the United States and in Asia. Many of them are holding prestigious academic positions and contributing to behavioral health sciences.

As Kexin's primary mentor in the doctoral program, I can strongly attest to Kexin's promise and preparedness to undertake the course of training and research proposed in her application. I will continue to mentor and collaborate with her throughout the period of the award and will use my experiences and expertise to help her in developing her long-term career goal of addressing the largely unstudied issue of living conditions on social isolation and the role of use of technology among older adults. In our specialized training Kexin will use findings from her project to develop technology related interventions to defend the negative impact of social isolation on cognitive impairments for older persons. In summary, Kexin is an enthusiastic social scientist who is capable and self-motivated. She produces high quality research output. Her good interpersonal skills will path the way for her future success. The support and training she would receive through this F99/K00 award will, I strongly believe, ensure her trail toward becoming an accomplished independent investigator and scientist. I enthusiastically offer my strongest possible recommendation for granting the fellowship award to Ms. Kexin Yu.

Yours sincerely,



Iris Chi, DSW
Chinese American Golden Age Association Frances Wu Chair Professor

USC Suzanne Dworak-Peck

School of Social Work

October 16, 2019

Re: Kexin Yu F99/K00 Application

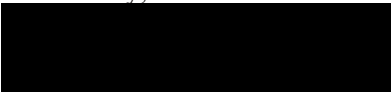
Dear Fellowship Review Committee,

I am delighted to serve as Ms. Kexin Yu's co-sponsor for her F99/K00 proposal for the F99 Phase. Ms. Yu has outstanding performance in her doctoral training program, and has worked closely with me on two of my research projects related to improve health and quality of life outcomes of older adults through technology adoption. The first project, Intergenerational Mobile Technology Opportunities Program, has three phases. The first phase was a demonstration project in Taiwan with 334 older adults with diabetes learning to use mobile technology and a diabetes self-care application to improve health behaviors and clinical and psychosocial outcomes. Kexin was involved in analyzing a subset of the data and publishing the results. Our research received the Nobuo Maeda International Research Award in 2018 from the American Public Health Association, Aging & Public Health Section. The second phase was a pilot study to understand feasibility and adaption of the IMTOP program to serve the Chinese American and Latino communities in Los Angeles. Kexin took the primary responsibility of recruiting older adult patients with diabetes, demonstrated the mobile applications, and conducted interviews and analyzed the qualitative and quantitative data. Her work has helped me in preparing an R18 dissemination and implementation proposal aiming to submit to the NIDDK. The third phase was a partnership with a senior housing social service agency to understand feasibility of implementing the IMTOP program in senior housing communities. Kexin joined the focus group discussion which helped her better understand how living environment affects older adults' social connectivity, loneliness, and health outcomes. The second project involved co-designing a mobile application with caregivers of older adults to help them improve their self-care behaviors. Kexin conducted needs assessment interviews, and testings of the application prototypes for improving its design and engagement.

These projects provide a foundation for the innovative and highly valuable research she proposes in her F99/K00 application. In her F99 phase, she will examine the effect of physical and social environments on social isolation and cognitive health using a mixed-methods approach, and explores the moderating effect of using information and community technology (ICT). In the K00 phase, she will examine the effectiveness of an ICT-enhanced conversational intervention on social isolation and its effect on cognitive health among older adults. These are impactful dissertation project and post-doctoral research that stand out to inform ways to improve social well-being and cognitive health for successful aging. Kexin has conducted a doctoral qualifying examination research in which she examined three waves of Health and Retirement Study data to disentangle the directionality and time sequence between the relationships of geriatric conditions and loneliness among older adults. If awarded, the F99/K00 grant will provide Kexin with a significant opportunity to extend her research for successful aging, and develop her expertise in the intersections of social relationships, ICT, and cognitive health research.

Kexin's intellectual focus and determination, methodological sophistication and writing skills, and unwavering commitment to strong and impactful science for the public good make her a highly promising independent investigator and scientist. I have successfully mentored many outstanding young scientists, and I enthusiastically offer my strongest possible recommendation for the F99/K00 fellowship award for Ms. Kexin Yu.

Sincerely,


Shinyi Wu, PhD Associate Professor
USC Suzanne Dworak-Peck School of Social Work
USC Edward R. Roybal Institute on Aging
USC Viterbi School of Engineering
Daniel J. Epstein Department of Industrial and Systems Engineering

USC Suzanne Dworak-Peck
School of Social Work

October 14, 2019

Re: Kexin Yu

NIA F99/K00 reviewers:

This letter affirms my enthusiastic support for Kexin Yu's F99/K00 application. I am the Albert G. and Frances Lomas Feldman Endowed Professor of Social Policy and Health in the Suzanne Dworak-Peck School of Social Work at the University of Southern California (USC). I also hold secondary appointments as professor in the departments of Anthropology and Preventive Medicine at USC. My expertise in implementation science, health disparities, mental health services for older adults, and mixed method designs, and my extensive track record of mentoring both National Institute of Health and National Science Foundation pre-dissertation and postdoctoral fellows, positions me as an ideal co-sponsor for the proposed award.

Kexin is an outstanding applicant for this award and it is my pleasure to serve as her co-sponsor. As a professor at the USC SSW who teaches mixed-methods/qualitative research methodology in the doctoral program, I have known Kexin since she took the class a year ago. Kexin is one of the most highly skilled and motivated graduate students I have worked with. She is a brilliant student whose research has demonstrated critical thinking, creativity, and competence to integrate qualitative and quantitative scientific approaches.

I am currently serving as a consultant to Dr. Dilip Jeste and colleagues at the IBM Center for Artificial Intelligence and Successful Aging at the University of California, San Diego on a mixed methods study examining the use of technology to address the problems of loneliness and isolation in older adults. I have served as principal investigator on multiple NIH funded research studies and have mentored numerous Ph.D. students across diverse disciplines, including public health, public policy, anthropology, and social work. My experiences have fully equipped me to work with Kexin on the mixed-methods inquiries of her dissertation project and at her postdoctoral training stage. We will meet once in a month to discuss the design, implementation and data analysis of the qualitative research with a purposive sample from the I-CONEC participants. The focus groups data collected in her dissertation stage will enable Kexin to contemplate the role of the living environment and technology usage on social isolation and cognitive health. With my expertise in dissemination and implementation research, I will be able to support Kexin in designing and developing a sustainable community-based intervention program for isolation and cognitive health in her postdoctoral training and after she takes on a faculty position.

I have reviewed Kexin's research and training plans. The dissertation and postdoctoral research agenda proposed will be an excellent opportunity to expand her research capacity and skills. Obtaining the F99/K00 funding and successfully achieving her specific aims will assist Kexin to become a future leader in aging research. I am committed to helping Kexin to complete the proposed study in a timely manner and will meet with her throughout her project to discuss her progress. It is my strong belief that Kexin's proposed dissertation and postdoctoral research will make important contributions to the knowledge of social determinants of isolation and cognitive health.

Sincerely,

Lawrence A. Palinkas, Ph.D.
Albert G. and Frances Lomas Feldman Professor of Social Policy and Health

University of Southern California



October 2, 2019



School of Medicine
Department of Neurology



Name of the fellowship applicant: Kexin Yu

Fellowship applicant's eRA Commons User Name: [REDACTED]

Funding Opportunity Announcement Number: RFA-AG-20-009

Dear review committee,

It is my pleasure to serve as Ms. Kexin Yu's co-sponsor for her F99/K00 application. I have witnessed the passion and dedication Ms. Yu has to the socially isolated older adult population through our collaboration thus far. Ms. Yu proactively made the connection between us one year ago at the Gerontological Society of Aging Annual Conference. I was impressed by her devotion in addressing the challenge of social isolation among older adults and our common research interests in employing technological tools to enhance social connections. To my great delight, Ms. Yu reached out to me when her research interests expanded to preventing cognitive decline among the high-risk population of older adults who lives in social isolation. I am very much looking forward to working with her and supporting her career development. If awarded, the F99/K00 grant will provide a significant opportunity for Ms. Yu to conduct self-directed research with data from my ongoing R01 projects (aka I-CONNECT project: R01AG056102, R01AG051828).

I have over 25 years of experience in dementia research and have mentored more than 10 postdoctoral fellows. Majority of the postdoctoral researchers were funded with NIH career development awards and successfully attained R01 or equivalent funding to support their independent research. I have no doubt that Ms. Yu will excel in her postdoctoral training and research as well as in further career stages as an emerging scholar. Through our collaboration made possible by the F99/K00 award, I will work with her during the entire period of the award, including the predoctoral and the postdoctoral stages. It is my plan to use my experience and expertise to facilitate Ms. Yu to achieve her long-term career goal of eradicating social isolation among older people and promote cognitive health exerting technology-facilitated social engagements.

At the pre-doctoral award stage, Ms. Yu will collaborate with me using qualitative research methods to explore sustainable community-based approaches to provide enduring social support to older adults at a low cost. This pilot effort is intended to serve as preliminary data for her to design and test the intervention program as a part of her independent research at the postdoctoral phase. Additionally Ms. Yu will join the data analysis and manuscript preparation of the I-CONNECT trial which provides her excellent learning and collaboration opportunities with the multi-disciplinary team of investigators. Since Ms. Yu is new to the field of cognitive health and dementia research, the Layton Aging and Alzheimer's Disease Center at Oregon Health and Science University will

be an outstanding training resource for her. I will direct Ms. Yu to different cores of the Layton center as her research projects require, such as the cores of outreach, recruitment and education, neuroimaging, and data management and analysis.

I strongly believe that Ms. Yu has the potential to become a leader in the research field of social isolation and cognitive health. This proposal makes a superb alliance between her current research activities and points to a clear and valuable direction of future research. Receiving this F99/K00 award will ensure her development as an accomplished independent scientist. I am thrilled to contribute to her training and aid her career development.

Sincerely,



Hiroko H. Dodge, PhD
Professor of Neurology
Director, Biostatistics and Data Core, NIA-Layton Aging and Alzheimer's Disease Center
Oregon Health and Science University, Portland, OR

Milton and Carolyn Kevreson Research Professor of Neurology
Director, Biostatistics and Data Core, Michigan Alzheimer's Disease Center
University of Michigan, Ann Arbor, MI

October 2019

MEMORANDUM

From: Michael Hurlburt, Ph.D. [REDACTED]
Associate Professor
Director of Doctoral Programs and Chair of the Ph.D. Program
Suzanne Dworak-Peck School of Social Work
University of Southern California (USC)

RE: **Kexin Yu, PhD Candidate at the University of Southern California, Suzanne Dworak-Peck School of Social Work**

Description of Institutional Environment & Commitment to Training

Dear Sir/Madame,

Social Work Ph.D. Program. The USC Suzanne Dworak-Peck School of Social Work (USCSSW) established the first social work doctoral program in the western United States in 1953. Today we continue the tradition of admitting highly motivated, self-directed individuals interested in university research and teaching. Students pursue an in-depth, customized course of study in an atmosphere of careful mentoring and respect for scholarship. Our doctoral curriculum is highly interdisciplinary with the intent of producing graduates who are capable of original research and passionate about advancing the profession's knowledge base. Our program fosters early engagement in research and publication, together with a structured and sequential experience in classroom teaching. Unique developmental opportunities are offered to improve students' presentation skills, networking with other universities, and leadership potential. Our goal is to make students competitive for the best available positions in this country and elsewhere in this world.

Program Structure. Doctoral students at USCSSW are required to complete a minimum of 45 course units beyond the master's degree (exclusive of the Doctoral Dissertation). During the first year of the PhD program, students complete 8 courses (16 units) in social work policy practice and research and participate in faculty-directed research projects. Core Substantive Courses include: Theories of Human Behavior in the Context of Social Environments; Explanatory Theories for Larger Social Systems; Theories for Practice with Small and Large Systems. Core Research/Statistics Courses include: Introduction to Social Work Statistics; Multiple Regression in Social Work Research; Advanced Social Work Research Methods. During the second year of the PhD Program, students are required to enroll in advanced multivariate statistics and at least three courses (per semester) in other departments or schools within the university. At least eight of these 12 units must be in courses with a substantive rather than a research methodology or statistics focus. Additional second year requirements include three student-directed tutorials (2-unit) under the mentorship of a social work faculty member. Tutorials provide greater understanding of the student's chosen specialty through a closer examination of relevant practice theories, explanatory theories, and research methodologies. Each tutorial is designed to produce a study of publishable quality. In the third year of the PhD Program, students must pass a qualifying examination in their research area, which consists of a written paper of publishable quality and includes an oral defense. After passing the qualifying exam, students are advanced to candidacy. Year Four and Beyond- Dissertation: Before proceeding with the dissertation, students must establish a dissertation committee approved by the Graduate School and submit a dissertation proposal for approval. Students submit their completed dissertation to their dissertation committee and orally defend it.

University of Southern California



Teaching. During their third and fourth year, doctoral students are required to teach one course per year. Teaching requirements may be fulfilled by co-teaching, teaching as an assistant or solo teaching. Before beginning these experiences, students must take a teaching course approved by the doctoral committee.

Average time to degree. The average time to degree over the past 10 years is 5.5 years.

Monitoring and evaluation. The USCSSW doctoral program requires an End of Semester Report from PhD students at the end of each semester for the entire duration of their doctoral studies. This report is completed by the student's faculty mentor and includes information on milestones and other deliverables (e.g., proposals, publications, presentations), as well as a description of the student's dissertation progress. This report is jointly signed by the student and mentor, and is submitted to the director of the doctoral program. The doctoral program also requires a Progress Report Timeline from PhD students at the end of each year of their doctoral studies. This Progress Report Timeline Report is used as a tool to guide and monitor each student's individual progress and ensures the PhD Program that each student is on track with specific tasks and milestones specific to each year of doctoral studies. This yearly report is completed by the PhD student's faculty mentor and is jointly signed by the student and faculty; this report is then submitted to the director of the doctoral program and to USC's graduate school department. These semester and yearly progress reports help doctoral students to navigate through the program productively and expeditiously.

Additional Program Opportunities and Commitment to Training. USCSSW and the University of Southern California are collectively committed to outstanding training, mentorship, and professional development for all of our doctoral students, including the applicant, Kexin Yu. Within USCSSW, the research clusters of excellence provide an environmental structure in which students have opportunities for discourse, collaboration, and professional training in the context of faculty, peers, and external collaborators who have broadly shared research interests. The research clusters regularly hold research seminars, which involve presentations and discussions of research by cluster members as well as by external speakers. Kexin will have opportunities through the research centers, particularly USC Edward R. Roybal Institute on Aging, to present and discuss her own work. Students also have strong collaborative connections with one another, which is facilitated by availability of dedicated office space for doctoral students at the Hamovitch Research Center office in downtown Los Angeles, and in a large, computer-equipped doctoral training lab in the School of Social Work on the main university campus. The Graduate School at USC provides additional resources for doctoral students, including specialized training in grant writing skills and other areas of professional development, which Kexin will take advantage of as part of her training plan. Students also have opportunities to engage in intellectual activities around the university, related to social science disciplines that Kexin will participate in. Finally, faculty within the USCSSW have strong connections with external colleagues and collaborators. With these strong and enduring ties, the applicant will have an easy ability to work with faculty outside of the SSW and to participate in the intellectual opportunities present at USC's Davis School of Gerontology, Keck School of Medicine and USC Alzheimer's Disease Research Center, such as seminars, courses, and ongoing lab involvement, as described extensively in the applicant's training plan. At all levels (institutional, school, and individual faculty), there is a strong commitment to the applicant's proposed training.

Kexin's dissertation chair and primary sponsor is Dr. Iris Chi, Dr. Frances Wu Chair for Chinese Elderly Professor, and distinguished scholar who has extensive experience conducting research focused on addressing health and mental health needs of older adults. Dr. Chi has more than 30 years of experience mentoring PhD students. She has been elected as a Fellow of the Gerontological Society of America, and the American Academy of Social Welfare and Social Work. All co-sponsors on the proposed project also possess strong records as recipients of NIH grants and are enthusiastic about the intellectual contributions Kexin brings to their ongoing research activities.

Progress/status of the F99/K00 applicant in relation to the program's timeline. Kexin is in excellent academic standing with respect to the timeline set forth by the USCSSW PhD program and anticipates completing her dissertation on time within standard degree completion. She has already completed all coursework, passed her qualifying exam, and advanced to PhD Candidacy. At the time of award notification, which will coincide with her 4th year in the program, Kexin will have defended her dissertation proposal. In her fourth year, Kexin will co-teach a doctoral-level statistical course. In her 5th years (Year 1 of the F99 award), Kexin will also continue her work with her lead sponsor, Dr. Iris Chi, as well as other faculty members. She will also work on projects that are designed to complement her research agenda and strengthen manuscript development, grant-writing, and grant management abilities.

As chair of the doctoral program, I am pleased to offer my full support for Kexin's F99/K00 application. Kexin has remarkable promise for a career in academic research and teaching, specifically in the area of dementia and social isolation intervention among older adults.

PHS Human Subjects and Clinical Trials Information

OMB Number: 0925-0001 and 0925-0002

Expiration Date: 03/31/2020

Are Human Subjects Involved

☒ Yes

☐ No

Is the Project Exempt from Federal regulations?

☐ Yes

☒ No

Exemption Number

☐ 1

☐ 2

☐ 3

☐ 4

☐ 5

☐ 6

☐ 7

☐ 8

Other Requested Information

Human Subject Studies

Study#	Study Title	Clinical Trial?
<u>1</u>	Examining the Longitudinal Influence of the Physical and Social Environments on Social Isolation and Cognitive Health: contextualizing the role of technology	No

Section 1 - Basic Information (Study 1)

OMB Number: 0925-0001 and 0925-0002

Expiration Date: 03/31/2020

1.1. Study Title *

Examining the Longitudinal Influence of the Physical and Social Environments on Social Isolation and Cognitive Health: contextualizing the role of technology

1.2. Is this study exempt from Federal Regulations *

☐ Yes ☒ No

1.3. Exemption Number

☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐ 6 ☐ 7 ☐ 8

1.4. Clinical Trial Questionnaire *

1.4.a. Does the study involve human participants?

☒ Yes ☐ No

1.4.b. Are the participants prospectively assigned to an intervention?

☐ Yes ☒ No

1.4.c. Is the study designed to evaluate the effect of the intervention on the participants?

☐ Yes ☒ No

1.4.d. Is the effect that will be evaluated a health-related biomedical or behavioral outcome?

☐ Yes ☒ No

1.5. Provide the ClinicalTrials.gov Identifier (e.g. NCT87654321) for this trial, if applicable

Section 2 - Study Population Characteristics (Study 1)

2.1. Conditions or Focus of Study

- Social isolation
- Cognitive functioning

2.2. Eligibility Criteria

Participants of the National Health & Aging Trends Study or participants of the Internet-based Conversational Engagement Clinical Trial (I-CONNECT)

2.3. Age Limits	Min Age: 65 Years	Max Age: N/A (No limit)
2.4. Inclusion of Women, Minorities, and Children	Att__4_Inclusion_of_WMC1016533165.pdf	
2.5. Recruitment and Retention Plan	Att__5_Recruitment_and_Retent_Plan1016533286.pdf	
2.6. Recruitment Status	Not yet recruiting	
2.7. Study Timeline	Attachment__6_Study_Timeline1016533224.pdf	
2.8. Enrollment of First Subject	05/01/2020	Anticipated

Inclusion of Women, Minorities, and Children

Inclusion of Women, Minorities

National Health and Aging Trends Study (NHATS) follows a nationally representative cohort of individuals aged 65 and above and enrolled in Medicare, with oversamples of the oldest-old age groups and non-Hispanic Black persons. At the baseline of the secondary data analysis with NHATS data (N=8,334), 59.12% were female, 20.52% identified as non-Hispanic Black, and 5.6% identified as Hispanic. These racial and gender statistics are largely representative of older adults across the nation.

The Internet-based Conversational Engagement Clinical Trial (I-CONNECT) is currently recruiting participants. Gender and racial break-down information is not available at this moment. This study does not exclude any individuals on the basis of race and/or gender. The participants of the focus group discussion will be sampled from the I-CONNECT participants. A purposeful sampling of qualitative research approach will be used to increase the probability of elicit diverse and rich information in the discussions. The anticipated enrollment for the focus groups has slightly more female participants than male participants, given women usually composite higher than 50% of the persons in older age groups. The purposeful sampling aims to recruit participants from racially diverse backgrounds.

Inclusion of Children

No children will participate in this study. All persons who participated in the NHATS surveys and I-CONNECT were above the age of 65.

Recruitment and Retention Plan

Focus group participants will be recruited from the participants of the I-CONNECT study. Working with the I-CONNECT research team, the PI or a research assistant will contact individuals who meet the purposeful sampling criteria via phone calls and invite them to participate in the focus group discussion. The study purpose, participants' rights and options will be carefully explained. If the individual agrees to participate, his/her availability will be asked and recorded in order to schedule the focus group sessions. Five days before the planned focus group session, the PI or the research assistant will seek confirmation of attendance from the participants. Because the qualitative focus group study does not have a longitudinal design, a retention plan is inapplicable in this case.

Study Timeline

Research activities	F99		K00			
	Year 0	Year 1	Year 1	Year 2	Year 3	Year 4
Submit IRB application	x					
Secondary data analysis using longitudinal NHATS data	x	x				
Consult with sponsors and cosponsors to prepare the interview guide for the qualitative study	x					
Recruit participants and conduct focus groups	x	x	x			
Transcribe and analyze the focus group data		x	x	x		
Prepare and submit manuscripts to peer-review journals	x	x	x	x	x	x
Present work at conferences	x	x	x	x	x	x
Conduct hands-on research with I-CONNECT data			x	x	x	x
Prepare to apply for an NIH career development award				x	x	
Co-design and develop a community based social isolation and cognitive health intervention with older adult participants, in consultation with sponsors			x	x	x	x

Inclusion Enrollment Reports

IER ID#	Enrollment Location Type	Enrollment Location
<u>Study 1, IER 1</u>	Domestic	Domestic

Inclusion Enrollment Report 1

Using an Existing Dataset or Resource* : ☒ Yes ☐ No

Enrollment Location Type* : ☒ Domestic ☐ Foreign

Enrollment Country(ies): USA: UNITED STATES

Enrollment Location(s): Domestic

Comments:

Planned

Racial Categories	Ethnic Categories				Total
	Not Hispanic or Latino		Hispanic or Latino		
	Female	Male	Female	Male	
American Indian/ Alaska Native	1	0	1	1	3
Asian	2	2	0	1	5
Native Hawaiian or Other Pacific Islander	1	1	0	0	2
Black or African American	3	2	1	1	7
White	4	2	2	1	9
More than One Race	1	1	2	2	6
Total	12	8	6	6	32

Cumulative (Actual)

Racial Categories	Ethnic Categories									Total
	Not Hispanic or Latino			Hispanic or Latino			Unknown/Not Reported Ethnicity			
	Female	Male	Unknown/ Not Reported	Female	Male	Unknown/ Not Reported	Female	Male	Unknown/ Not Reported	
American Indian/ Alaska Native	0	0	0	0	0	0	0	0	0	0
Asian	0	0	0	0	0	0	0	0	0	0
Native Hawaiian or Other Pacific Islander	0	0	0	0	0	0	0	0	0	0
Black or African American	1064	646	0	0	0	0	0	0	0	1710
White	3351	2352	0	0	0	0	0	0	0	5703
More than One Race	3	5	0	0	0	0	0	0	0	8
Unknown or Not Reported	128	116	0	266	201	0	115	87	0	913
Total	4546	3119	0	266	201	0	115	87	0	8334

Section 3 - Protection and Monitoring Plans (Study 1)

3.1. Protection of Human Subjects

Att__8_Protection_of_Human_Subjects1016533225.pdf

3.2. Is this a multi-site study that will use the same protocol to conduct non-exempt human subjects research at more than one domestic site?

☐ Yes ☒ No ☐ N/A

If yes, describe the single IRB plan

3.3. Data and Safety Monitoring Plan

3.4. Will a Data and Safety Monitoring Board be appointed for this study?

☐ Yes ☒ No

3.5. Overall structure of the study team

Protection of Human Subjects

Risk to Human Subjects

a. Human Subjects Involvement, Characteristics, and Design

The proposed study involves both primary qualitative data analysis and secondary quantitative data analysis to examine the influence of the physical and social living environment, ICT use within the context of the environment, on social isolation and cognitive health. Secondary data analysis for the dissertation work will use wave 5 to 8 data from the National Health & Aging Trends Study (NHATS) (U01AG032947). A cohort of 8,334 individuals aged 65 and above responded to the wave 5 NHATS survey and received follow-up survey interviews annually after. Secondary quantitative data analysis at the proposed postdoctoral stage study will use data from the Internet-based Conversational Engagement Clinical Trial (I-CONNECT) study (NIA R01AG056102, R01AG056628 & R56AG051628). I-CONNECT project is currently recruiting socially isolated adults, aged 75 and above, and with normal cognition or mild cognitive impairment (MCI), to participate in the trial. I-CONNECT is a multi-site study conducted at both the University of Michigan (UM) and Oregon Health and Science University (OHSU). The recruiting goal is 160 participants at each site, 320 participants in total. As of October of 2019, 105 participants have enrolled in the study. The participants of the focus group discussion on the living environment, social isolation, and cognition will be purposefully sampled from the I-CONNECT participants with an intention to recruit individuals residing in varied living environments and with diverse demographic backgrounds. Two focus groups will be held at each site with a target of 8 participants in each focus group, resulting in 32 participants for the focus groups anticipated. The secondary data analysis procedures will be submitted to the University of Southern California's (USC) Institutional Review Board (IRB). The primary data collection and analysis procedure will be submitted to the IRB at USC, OHSU, and UM. I-CONNECT research team members at OHSU and UM will help with the participant recruitment for the focus groups and coordinate to book spaces for the focus groups.

b. Study Procedures and Materials

The secondary data analysis for the proposed dissertation study will use longitudinal data from the NHATS study. The proposed postdoctoral research will use data from I-CONNECT. NHATS data survey data were collected with in-person interviews using surveys with detailed data collection instructions designed by NHATS team researchers. I-CONNECT is currently recruiting participants from the communities around UM and OHSU, and randomizing enrolled participants into the video-chat intervention group and control group. The intervention group will have a computer installed in their homes with internet service. During 6-months long high dose duration, the participants assigned to the intervention group will have a video chat with study staff for 30 minutes/day, 4 times/week. Afterward, the participants will have 2 times/week video chats, 30 minutes/day, for an additional 6-months maintenance dose period. Both intervention and control group participants will have a 10-minutes telephone check-in with the study staff, answering brief questions to monitor their social activities and health conditions. At baseline, 6-month and 12 months, research staff will visit the participants at their homes to administer assessments in-person. Participants at both study sites will have MRIs at baseline and 6 months if receiving MRIs were safe for them. All video chat and in-home assessment sessions will be recorded for speech and language analysis. Only participants at OHSU will have their medication compliance tracked with electronic pillboxes.

Focus groups participants will be recruited from both intervention and control groups of the I-CONNECT participants. At each study site, two focus groups will be held – one with intervention group participants, the other with control group participants. The focus group sessions will be led by the PI and coordinated by a research assistant who will be taking notes for the discussion sessions. Participants will be notified that the focus groups will be audio recorded when scheduling the focus group sessions with them via phone calls. The PI/research assistant will explain to the participants the audio recording and notes will be used for research purpose only. The purposes of the focus group study, discussion topics, time frame, and incentive for data collection will be explicit in the informed consent form. Each focus group session will last for about an hour, at a selected meeting room that allows or privacy where the participants will not be seen or overheard by someone not participating in the focus groups. All participants will be advised to keep the discussion to the group and respect each other's opinions. The recordings of the focus groups will be transcribed verbatim for the use of data analysis.

c. Potential Risks

The risk of participating in the quantitative and qualitative components of the proposed study will be minimal. All primary and secondary data is going to be analyzed will be de-identified. The PI has no access to the names or any identifying data for the NHATS dataset. In order to purposefully sample participants for the focus group discussion, the group assignment, age, gender, race, and location will be used. Participants' contact information will be obtained to contact participants via phone calls. The contact information will be stored separately in a

password-protected file. Under no circumstances, the participant ID will be put together with the name and participant ID in the same file. Transportations from and back to their homes will be arranged for the participants to attend the focus group sessions. During the focus group discussion, participants might experience mild discomfort due to the discussion of topics related to social isolation. It will be explained to the participants they have the right not to answer any questions they do not wish to reply to. Participants will be informed they can withdraw from the study at any time and be compensated for their time and effort even if they cannot complete the focus group discussion. All data will be analyzed and presented in manuscripts anonymously.

Adequacy of Protection Against Risks

a. Informed Consent and Assent

The I-CONNECT participants signed informed consent for the parent study. Additional informed consent will be sought for the qualitative focus groups. To avoid contamination with the intervention effect, only participants who have completed the I-CONNECT study will be invited for the focus groups. Participants will be first contacted via phone calls. The PI/research assistant will describe the focus group procedures, including the plan of audio-recording, to the participants. If the individual agrees to participate, the PI/research assistant will schedule the time of the focus group with him/her. At the time of the focus groups, before any data would be collected, the PI or a research assistant will present written informed consent to the participants, answer any questions they might have, and then have the participants sign the informed consent. Individuals with MCI can provide their informed consent. The informed consent forms are going to be stored in a locked file cabinet for three years after completion of the study unless IRB deemed the requirement of the informed consent could be waived.

b. Protection Against Risk

All secondary data will be cleaned and recoded prior to being accessed by the PI, which reduces the risk of confidentiality. Primary qualitative focus group transcripts and notes will be stored in password-protected files. In the data collection process, participants will be informed that their participation in both the I-CONNECT and the qualitative focus groups are voluntary. They have the right to withdraw at any point in the study without any penalty. The data collected are confidential and will only be used for research-related purposes. In the focus group discussions, participants will be addressed with their first name. Their last name will not be asked or recorded. Only the research team and the Human Subjects Protection Program will have access to the primary raw data. Any research assistant who will help to contact participants, take note in the focus group discussion, to help with transcribing the audio recording, and/or to help with data analysis will receive IRB training and been added to the IRB record as research personnel prior to any research-related activities. In an appropriate time after the completion of data analysis, the original focus group summaries will be destroyed. The results from the analysis of the primary and secondary data will be published or presented at conferences in an anonymous/aggregated form. No identifiable information will be used.

Potential Benefits of Proposed Research to Human Subjects and Others

Secondary analysis using the NHATS data will contribute to the literature with longitudinal empirical evidence on the influences of physical and social environments on social isolation and cognitive health. Participants of the NHATS study might benefit from participating in the study as the evidence built from the data will be used to inform the development of interventions and policy related to aging friendly communities and aging in place. The I-CONNECT study is one of the first clinical trials examining the effectiveness of internet facilitated video chat intervention on reducing social isolation and prevent cognitive decline. Participants could directly benefit from the potential positive effects of the intervention. Knowledge will be built using the data collected in the I-CONNECT could benefit the participants indirectly by explaining one of the mechanisms through which internet-enhanced social interaction promote older adults' wellbeing and cognitive health. Participating in the focus group discussion could alleviate the experience of social isolation among the participants. Participants of the focus group will be engaged in a conversation exploring the design for a sustainable community-based program for isolation and cognitive health, which could indirectly benefit the participants by informing the development of an intervention.

Importance of the Knowledge to be Gained

Knowledge built in the proposed dissertation and postdoctoral research could lead to the development of a sustainable community-based ICT-facilitated program with the aim to address social isolation and cognitive decline. Risks to participation will be minimal. Knowledge to be gained is of extreme importance as the accelerated population aging poses significant challenges in providing health care and social services that meet the needs of the older adult population. The proposed research has the potential to inform the policy and practice regarding aging in place, to promote cognitive health, and to aid older adults to live longer independent lives.

Section 4 - Protocol Synopsis (Study 1)

4.1. Brief Summary

4.2. Study Design

4.2.a. Narrative Study Description

4.2.b. Primary Purpose

4.2.c. Interventions

Type	Name	Description
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4.2.d. Study Phase

Is this an NIH-defined Phase III Clinical Trial? ☐ Yes ☐ No

4.2.e. Intervention Model

4.2.f. Masking ☐ Yes ☐ No

☐ Participant ☐ Care Provider ☐ Investigator ☐ Outcomes Assessor

4.2.g. Allocation

4.3. Outcome Measures

Type	Name	Time Frame	Brief Description
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4.4. Statistical Design and Power

4.5. Subject Participation Duration

4.6. Will the study use an FDA-regulated intervention? ☐ Yes ☐ No

4.6.a. If yes, describe the availability of Investigational Product (IP) and Investigational New Drug (IND)/ Investigational Device Exemption (IDE) status

4.7. Dissemination Plan

Delayed Onset Studies

Delayed Onset Study#	Study Title	Anticipated Clinical Trial?	Justification
The form does not have any delayed onset studies			

Authentication of Key Biologicals and Chemical Resources

No key biological and/or chemical resources will be used in the activities proposed in this application.